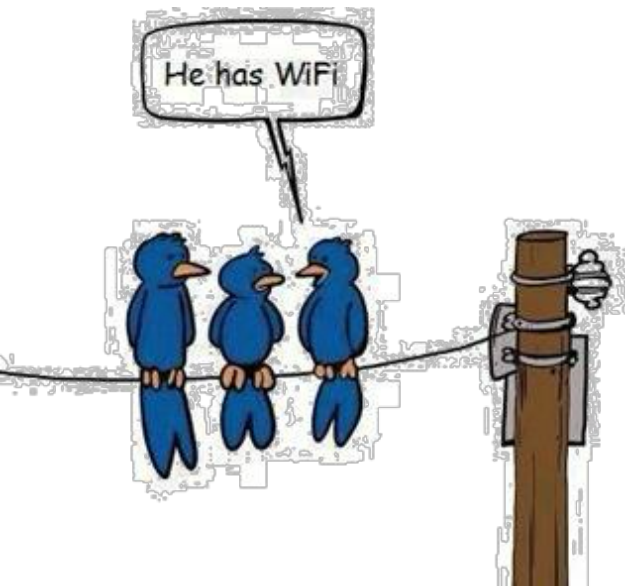


ECSE 4560/ 6560

Modern Communication Systems

Lecture 21: Wireless Sensing Part 2: Localization

Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Localization**
- Projects
 - Group formation [MCS Fall25 project.xlsx](#)
 - Project mid-term report due **Monday 11/10**
 - Final Project Presentation and Demos: Dec 9 and Dec 12
 - Project Report and peer evaluation: Finals exam week
- HW/Exams Dates
 - HW5 (MIMO Theory) due **Mon 11/10**. Late due (Wed 11/13)
 - HW6 (MIMO code) due **Fri 11/14**
 - HW7 (Sensing Theory) due **Fri 11/21** – covered in Exam 2
 - Exam 2: **Friday Nov 21** - Anyone needs makeup exam?

Mon-

Post today

Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae
 - Parts of Chapter 1,2,3,4,7
 - **MIMO Chapter 6**
- Sensing Readings
 - Xiong, Jie, and Kyle Jamieson. "ArrayTrack: A Fine-Grained indoor location system." *NSDI 2013*
 - 2024 SIGMOBILE Test-of-Time Award 

Today's Class

- MIMO Review (HW 6)
- Clock Synchronization in MIMO
- Intro-Wireless Sensing
 - Localization Techniques

Today's Class

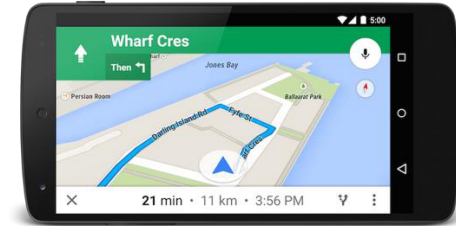
- MIMO Review (HW 6)
- Clock Synchronization in MIMO
- **Intro-Wireless Sensing**
 - **Localization Techniques**

Wireless Localization / Positioning

The process of obtaining a human or object's location using wireless signals

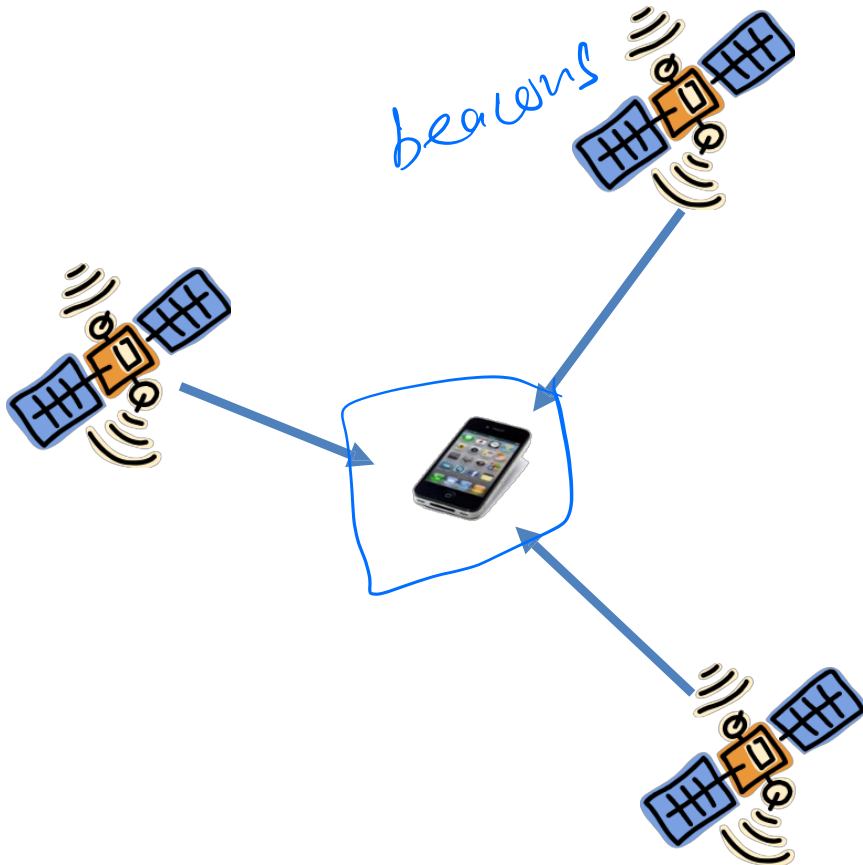
Applications:

- Navigation: outdoors (GPS) and indoors (e.g., museum)
- Location based services: Tagging, Reminder, Ads
- Virtual Reality and Motion Capture
- Gestures, writing in the air
- Behavioral Analytics (Health, activities, etc.)
- Locating misplaced items (keys)
- Location based security
- Delivery drones

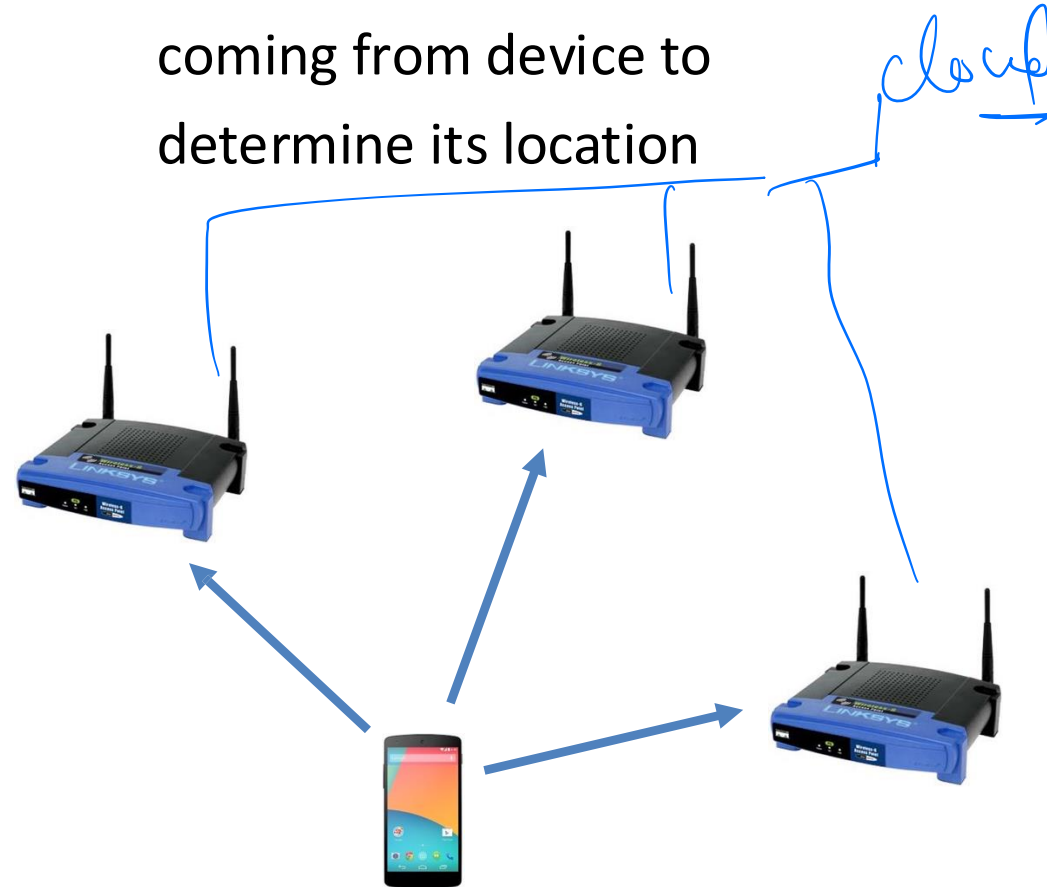


Wireless Localization Architecture.

- Device based: A device uses incoming signal from one or more “anchors” to determine its own location

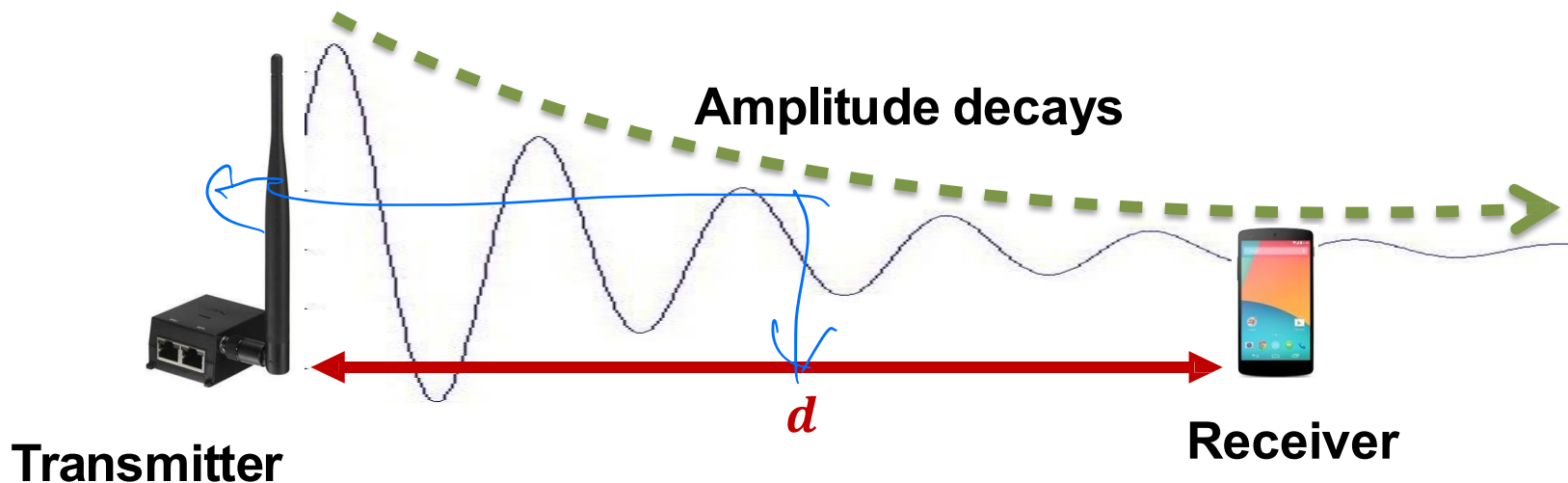


- Network based: Anchors (or Access points) use the signal coming from device to determine its location



Method 1: RSSI Based Localization

- Higher received power → Closer
- Lower received power → Farther

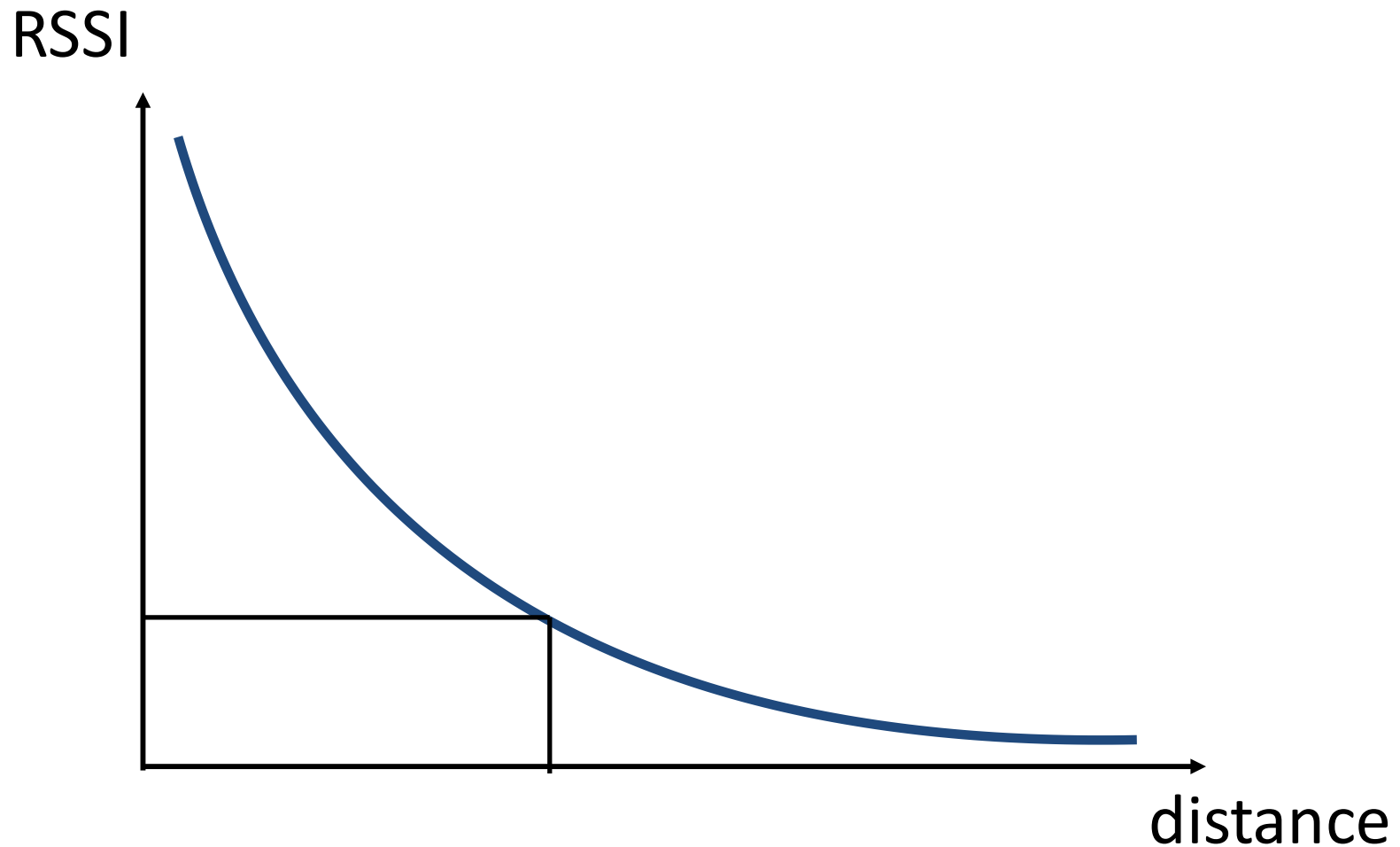


$$P_{Rx} = \frac{G_{Tx} G_{Rx} \lambda^2}{(4\pi d)^2} P_{Tx} \quad \Rightarrow \quad RSSI \propto \frac{1}{d^2}$$

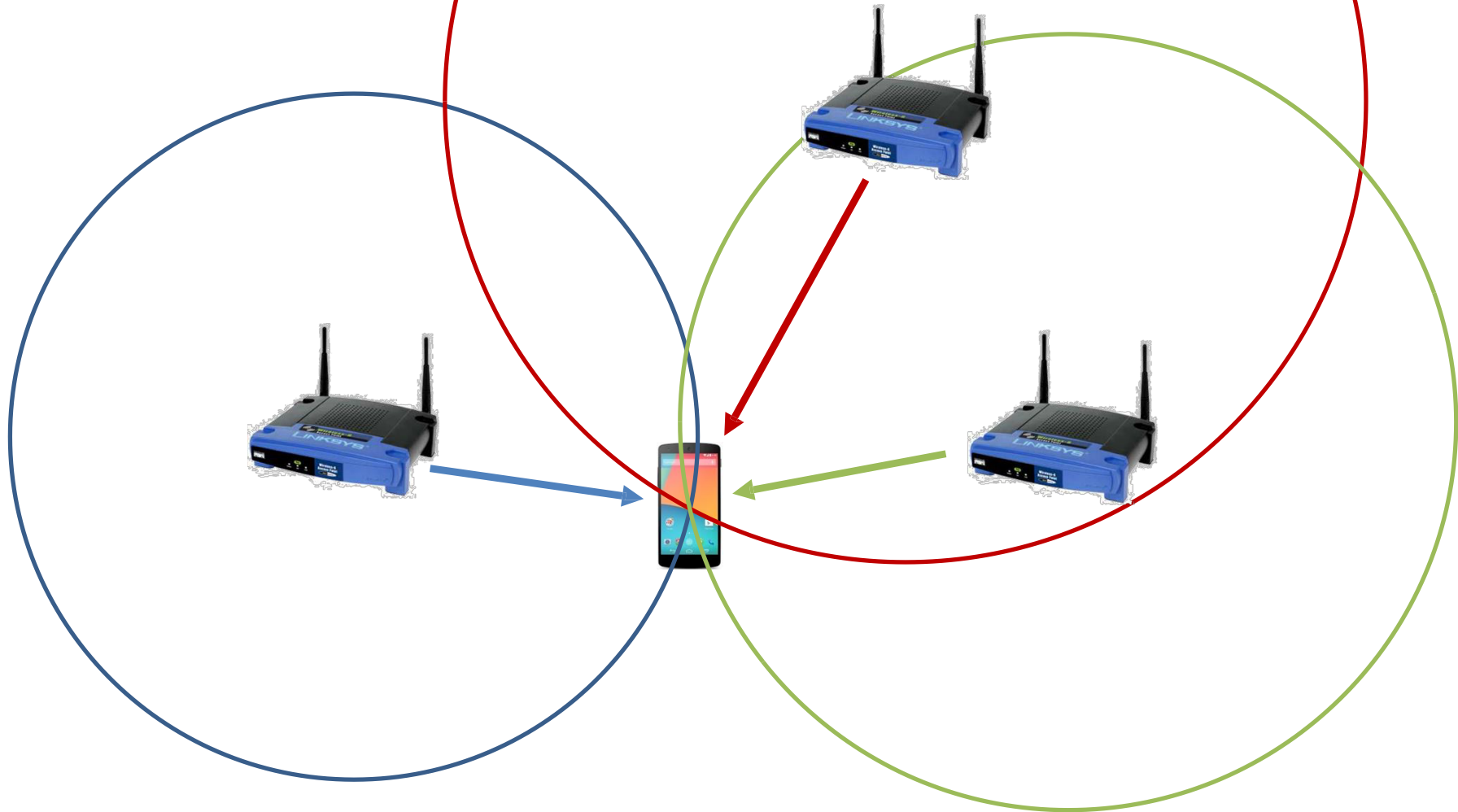
Use RSSI to estimate distance from APs!

Method 1: RSSI Based Localization

$$RSSI \propto \frac{1}{d^2}$$



Method 1: RSSI Based Localization

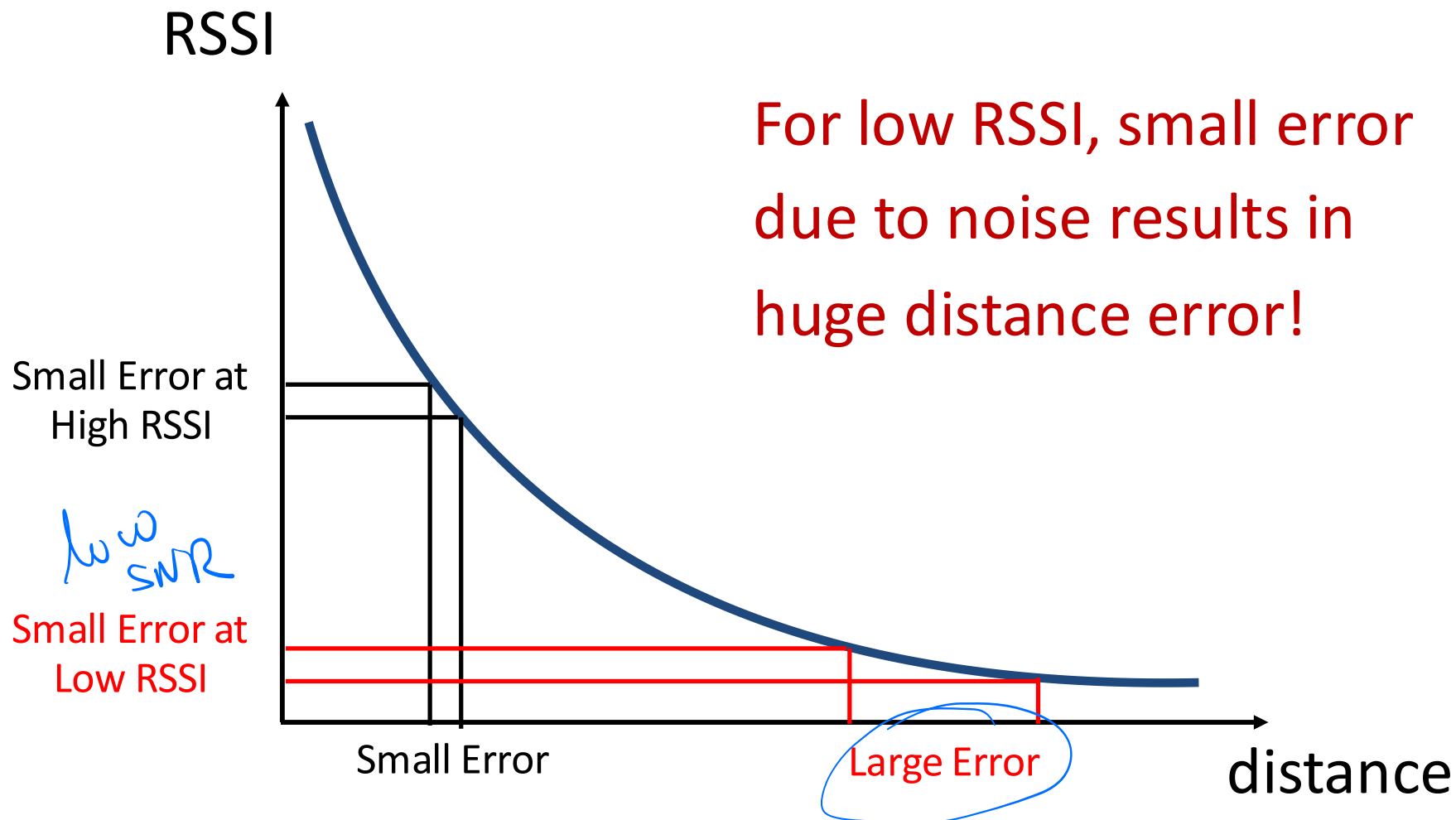


Trilateration

Method 1: RSSI Based Localization

Pros: Very simple, no hardware modifications

Cons: Highly inaccurate!



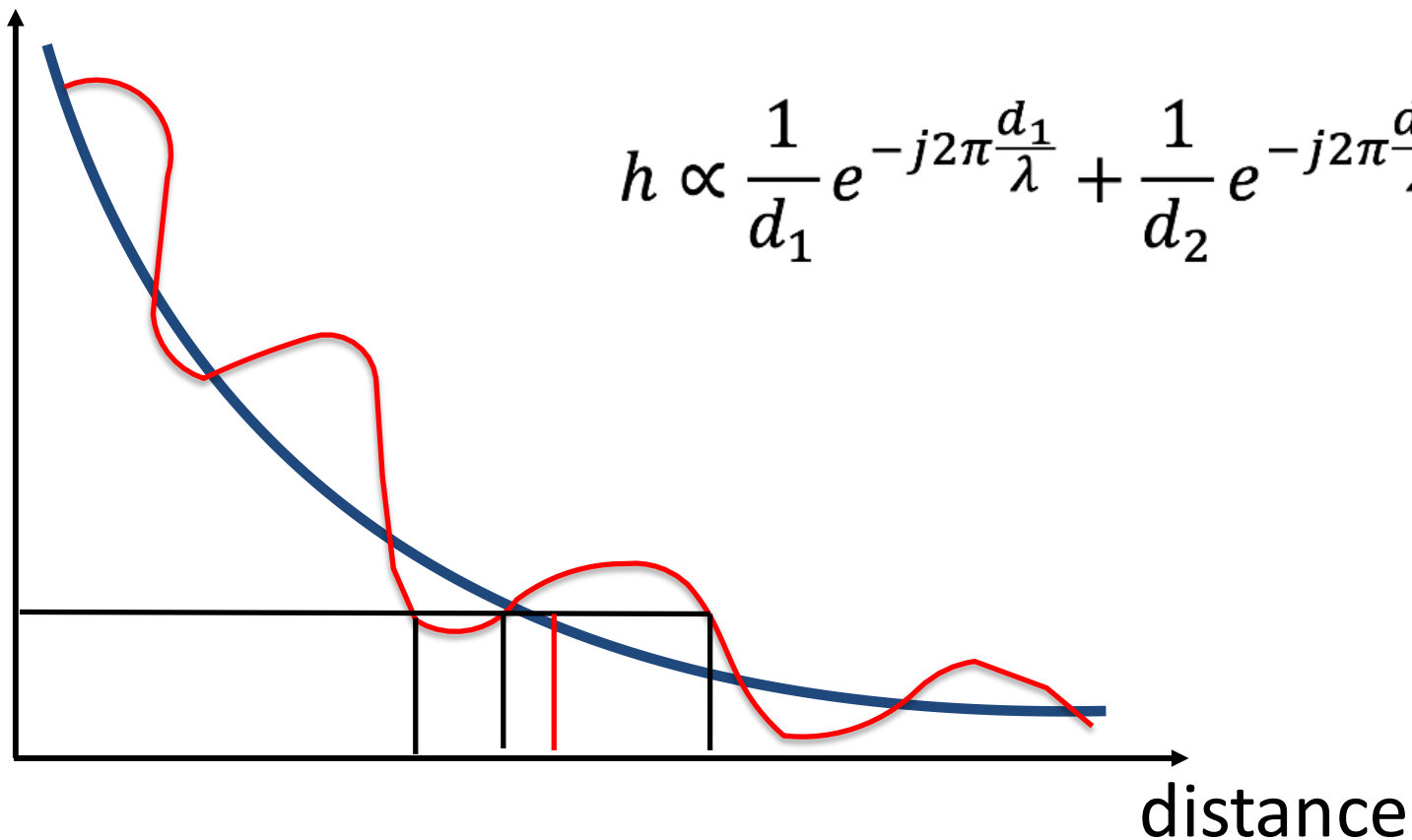
Method 1: RSSI Based Localization

Pros: Very simple, no hardware modifications!

Cons: Highly inaccurate!

Does not work with multipath! & blockage

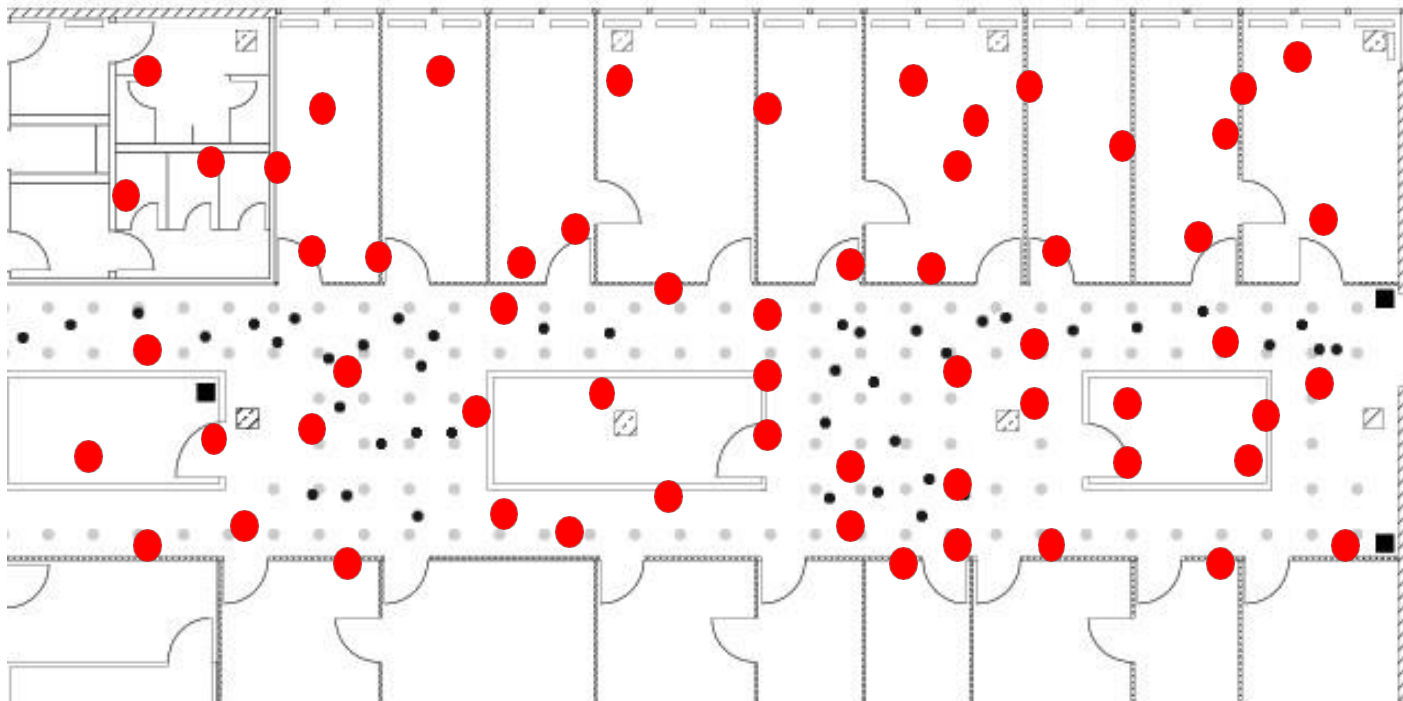
RSSI



Method 1: RSSI Based Localization

Solution: Fingerprinting → m, l

Measure and records RSSI fingerprints at each location (war-driving)

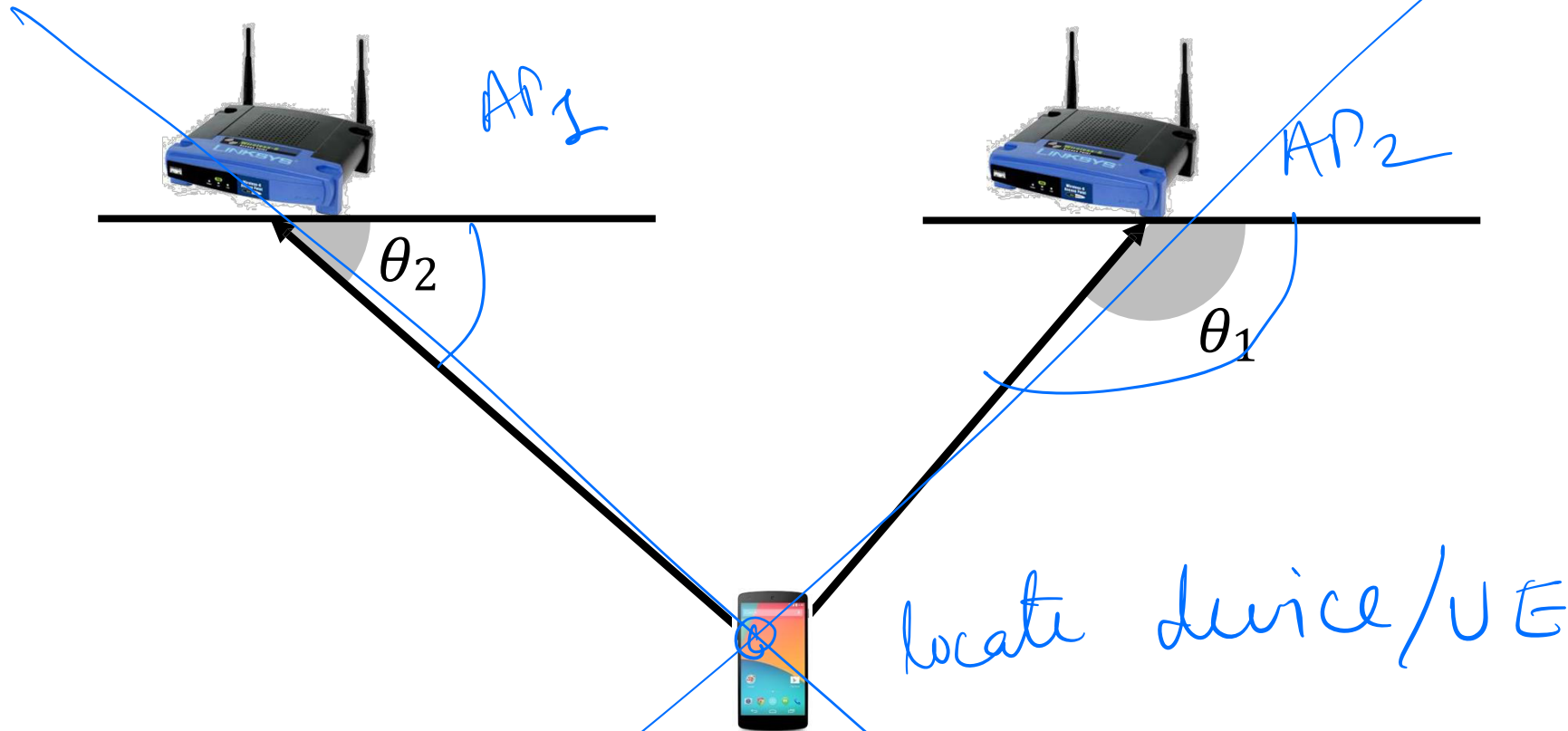


Pros: Works with multipath, No need to know AP locations!

Cons: Changes in environment/movement → change RSSI!
Continuous training is needed. Lots of effort!

Method 2: AoA Based Localization

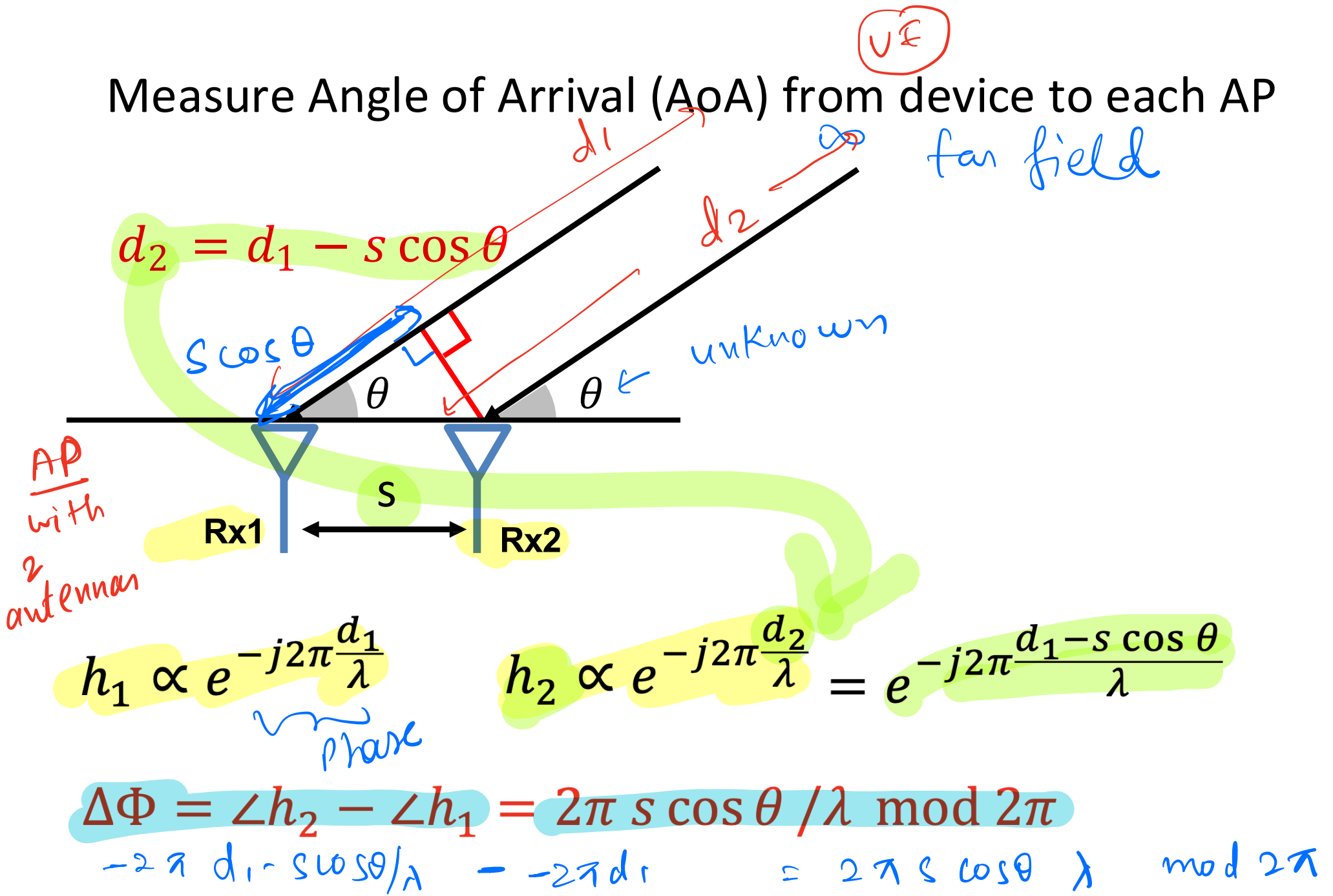
Measure Angle of Arrival (AoA) from device to each AP



Triangulation

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



$$\Delta\phi = 2\pi \frac{s \cos\theta}{\lambda} \pmod{2\pi} \quad + \text{noise}$$

$$\cos\theta = \frac{\Delta\phi}{s} \frac{1}{2\pi}$$

$$\hat{\theta} = \cos^{-1} \left(\frac{\Delta\phi}{s} \frac{1}{2\pi} \right) \quad + \text{noise}$$

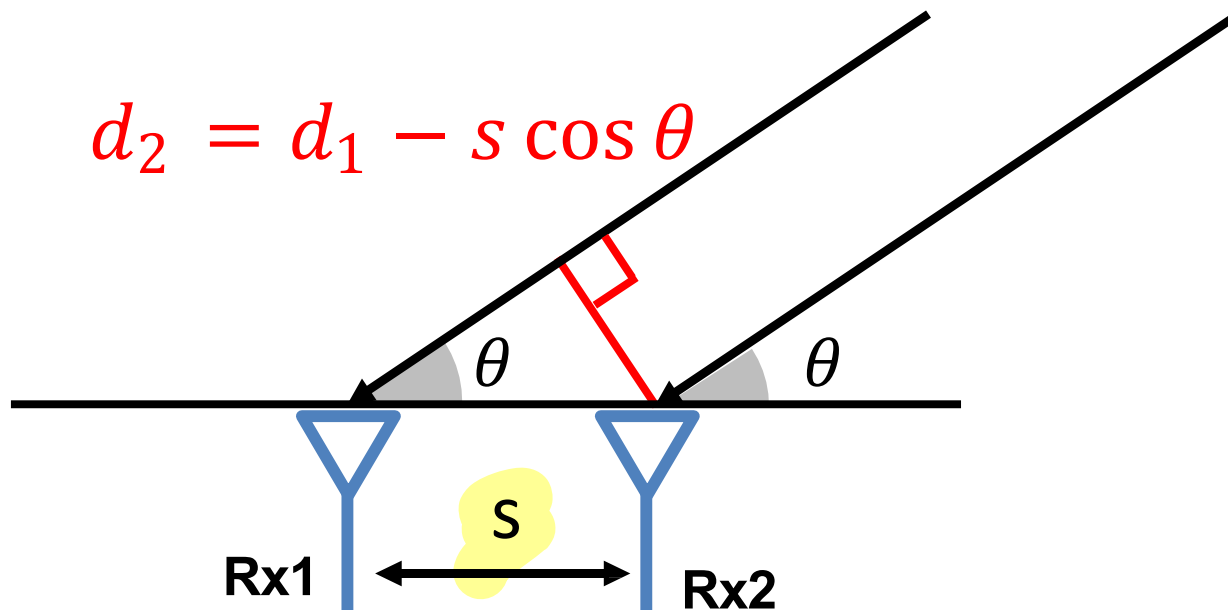
two possible estimates

$$\hat{\theta} \text{ \& \; } -\hat{\theta}$$

() (N)

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



$$\Delta\Phi = \angle h_2 - \angle h_1 = (2\pi s \cos \theta / \lambda) \bmod 2\pi$$

Ambiguity: $\exists \theta_1 \neq \theta_2 \mid \Delta\Phi_1 = \Delta\Phi_2 \bmod 2\pi$

To avoid ambiguity, we want: $-\pi \leq \Delta\Phi \leq \pi$

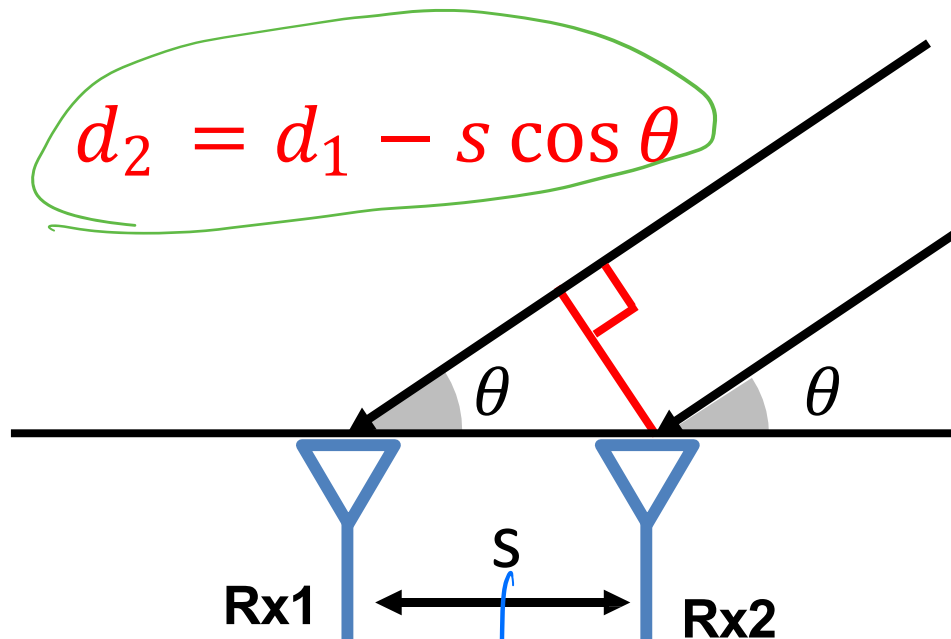
alternate way

$$0 \leq \Delta\Phi \leq 2\pi$$

unique value

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



We want:

$$-\pi \leq \Delta\Phi \leq \pi$$

to avoid ambiguity

We have:

$$-1 \leq \cos \theta \leq 1$$

$$-2\pi \frac{s}{\lambda} \leq 2\pi \frac{s}{\lambda} \cos \theta \leq 2\pi \frac{s}{\lambda}$$

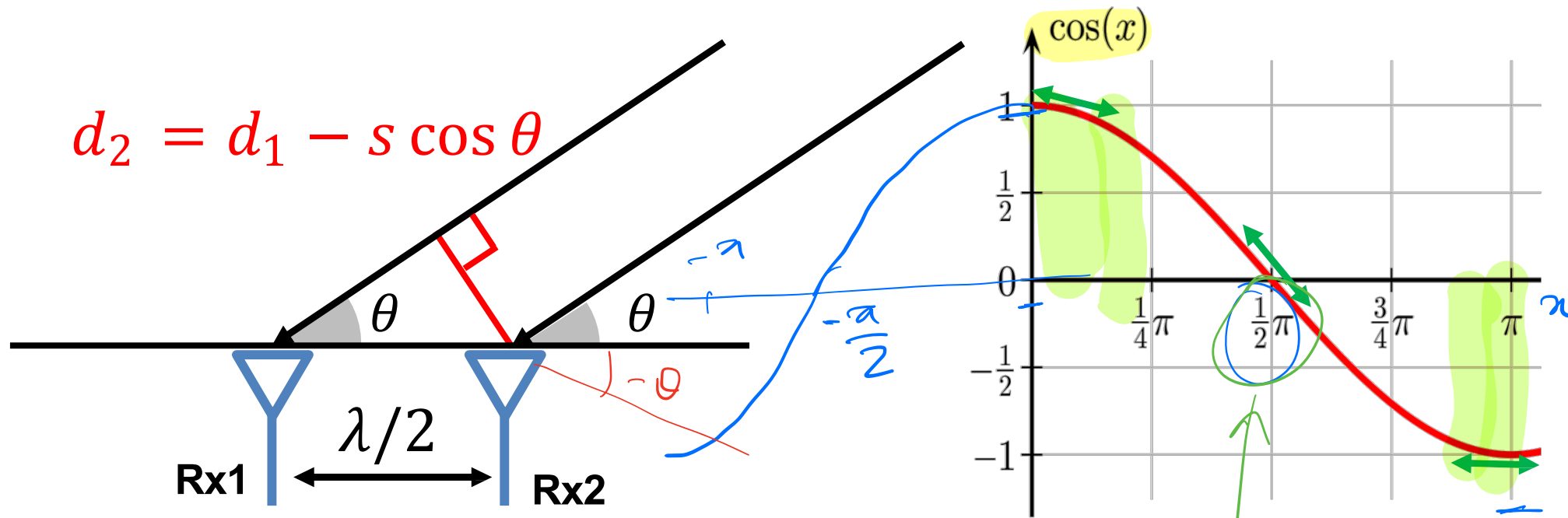
$$-2\pi \frac{s}{\lambda} \leq \Delta\Phi \leq 2\pi \frac{s}{\lambda}$$

$$-\pi \leq \Delta\Phi \leq \pi$$

Set: $s = \lambda/2$

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



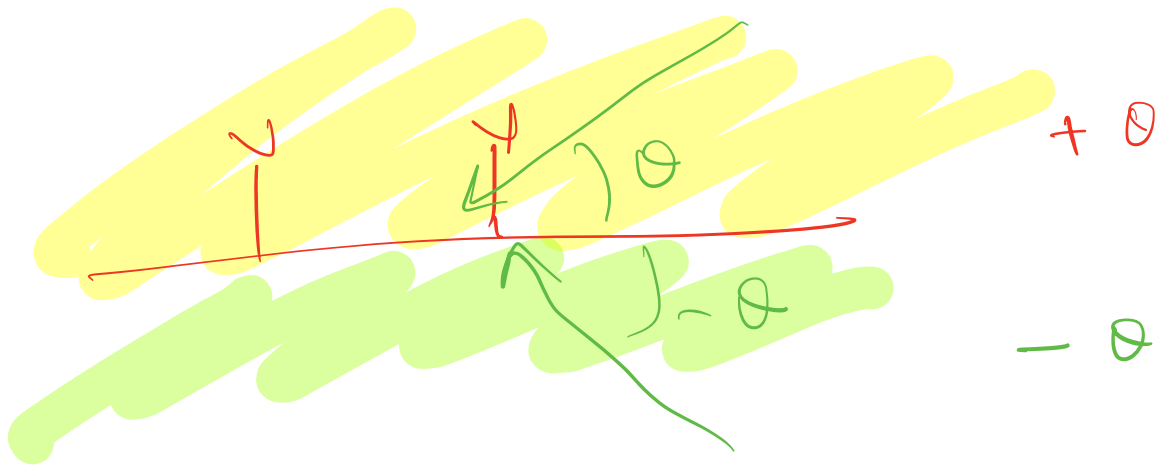
Pros: More accurate than RSSI, Simple!

Cons: Ambiguity: $\cos \theta = \cos(-\theta)$

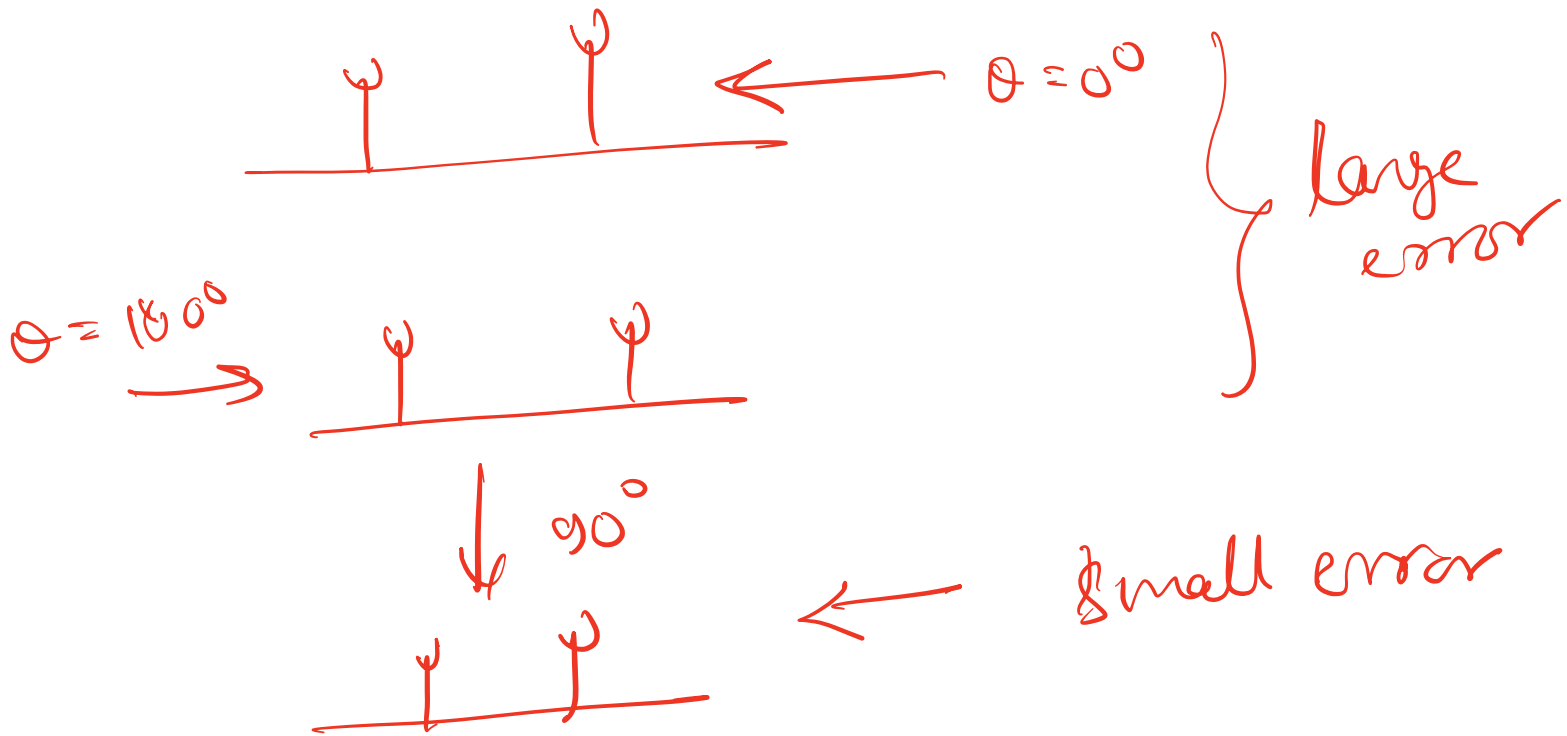
Error not linear with θ due to $\cos \theta$

Requires 2 Antennas separated $\lambda/2$

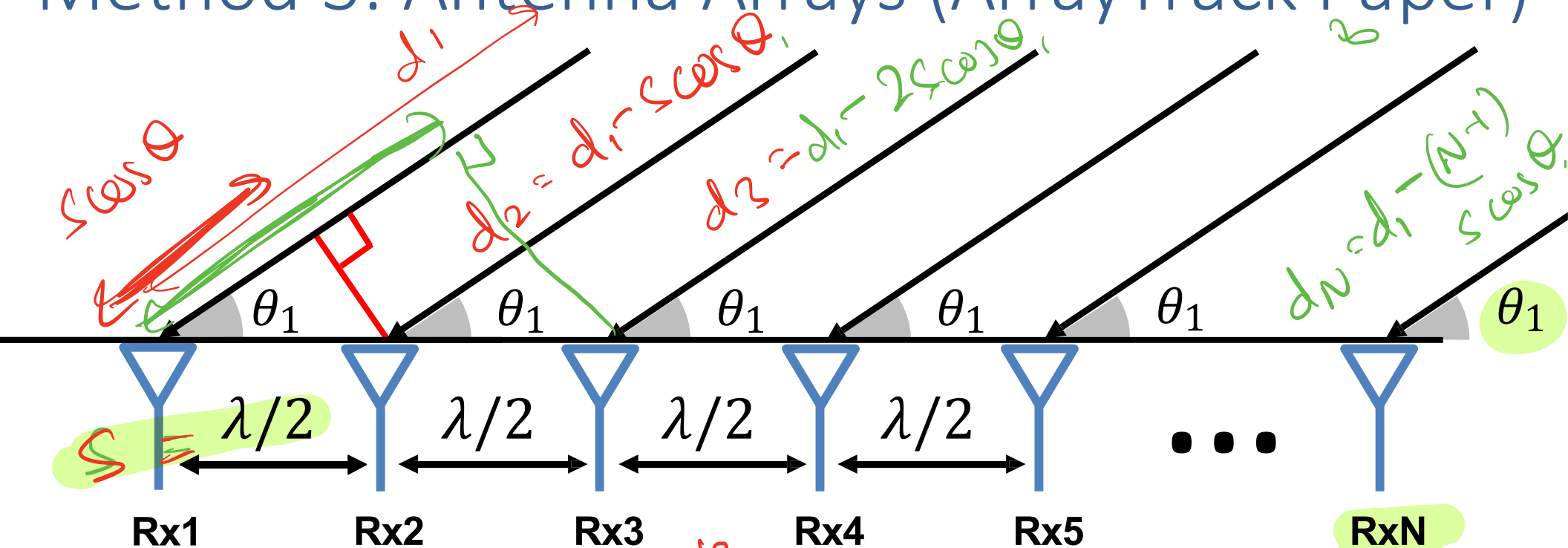
Does not work with multipath!



error is large at $\theta \sim 0^\circ$
 or $\theta \sim 180^\circ$



Method 3: Antenna Arrays (ArrayTrack Paper)



$$h_k = \alpha_1 e^{-j2\pi \frac{d_1 - k s \cos \theta_1}{\lambda}} = \alpha_1 e^{-j\phi_1 + k\pi \cos \theta_1}$$

Handwritten notes: α_k , ϕ_1 , N antennas.

$\alpha_1, \alpha_2, \dots, \alpha_N$

$k = 0$ for 1st antenna
 $k = N-1$ for N th antenna

$$\boxed{2\pi S/A = 2\pi \frac{A}{2\pi} = \pi}$$

$$\begin{aligned} & \begin{bmatrix} h_1 \\ h_2 \propto h_1 e^{j\pi \cos \theta_1} \text{ + noise} \\ h_3 \propto h_1 e^{j2\pi \cos \theta_1} \text{ + noise} \\ \vdots \\ h_K \propto h_1 e^{j(K-1)\pi \cos \theta_1} \\ \vdots \\ h_N \propto h_1 e^{j(N-1)\pi \cos \theta_1} \end{bmatrix} \end{aligned}$$

to estimate θ_1 $\angle h_2 - \angle h_1 = \pi \cos \theta_1$

use $h_1 \dots h_N$ to estimate θ_1

FFT-based technique

Take FFT across $h_1, h_2, \dots, h_N$

spatial-FFT

$$B(\psi) = \text{FFT} \left(\bar{h}(k) \right)$$

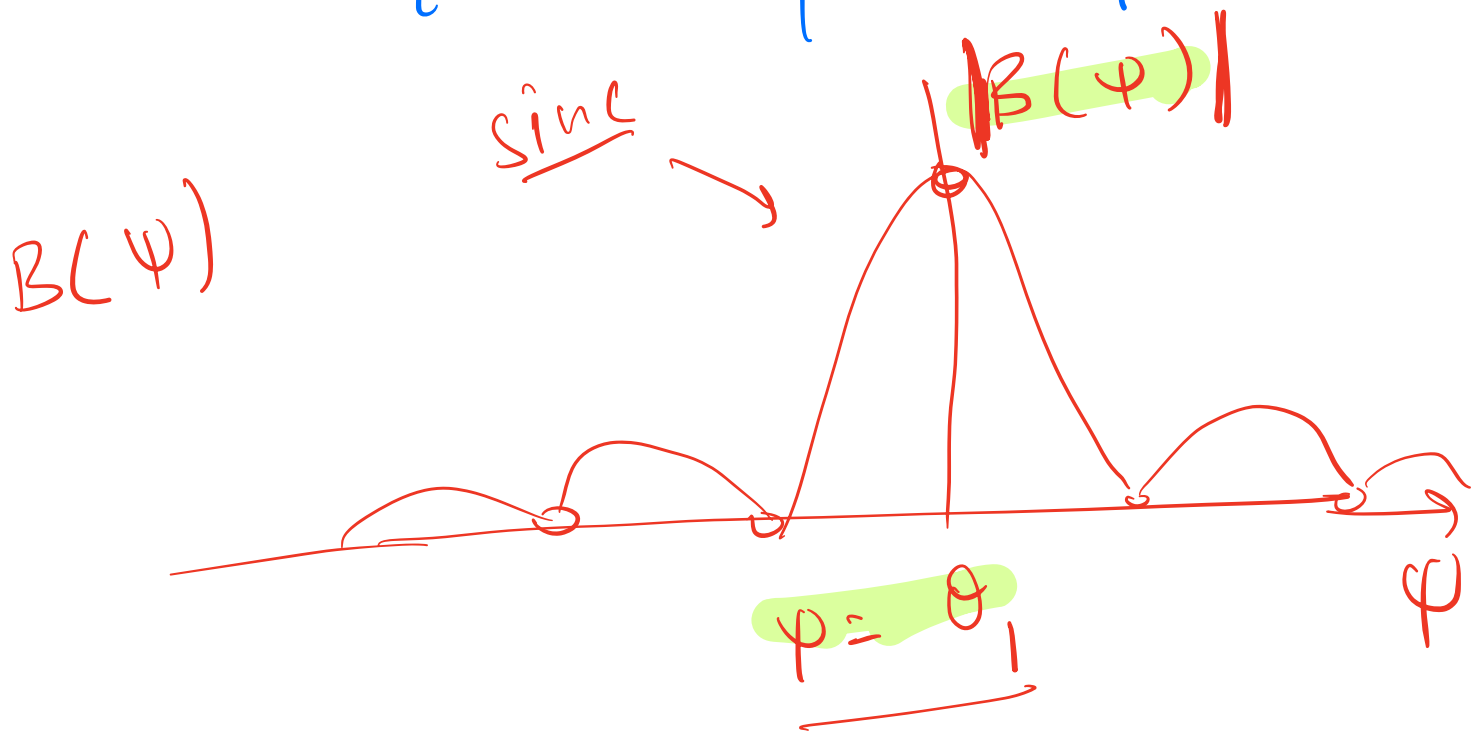
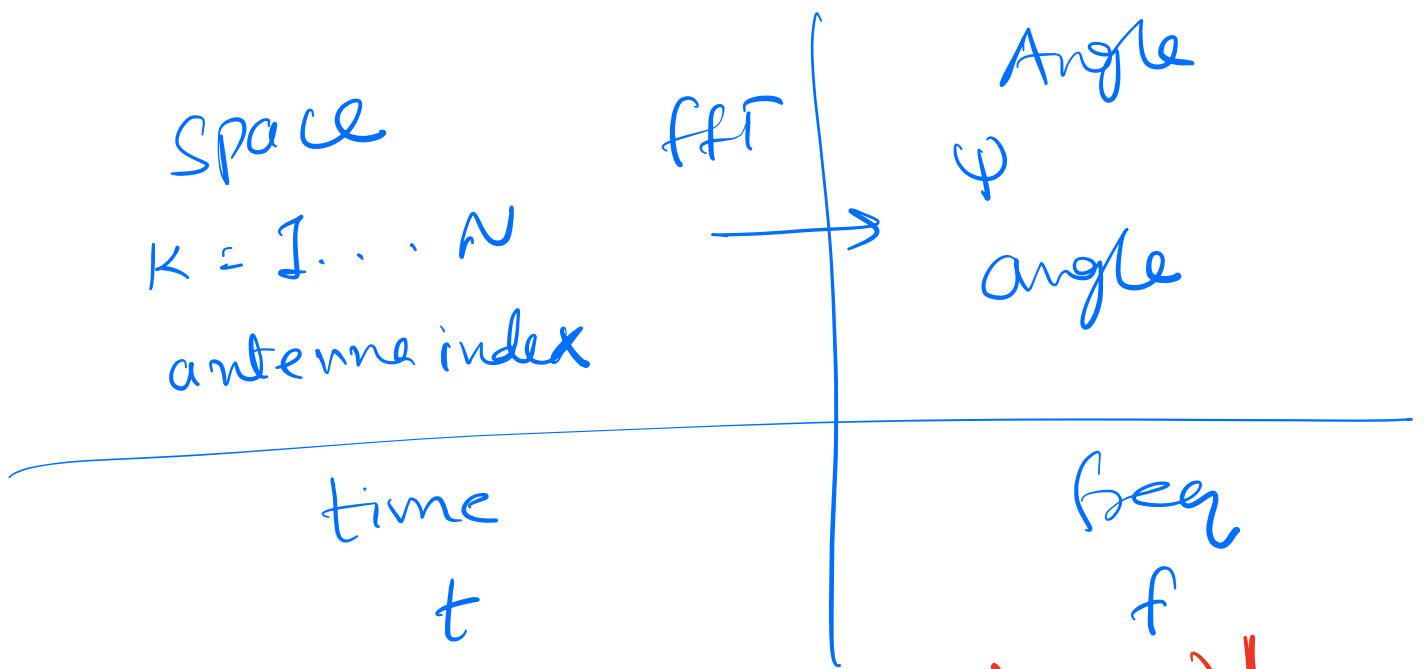
$$h_N(k) = h_1 e^{j \underbrace{K \pi \cos \theta_1}_{\text{freq}}}$$

time $\uparrow$ freq $\uparrow$

at freq $\cos \psi = \cos \theta_1$

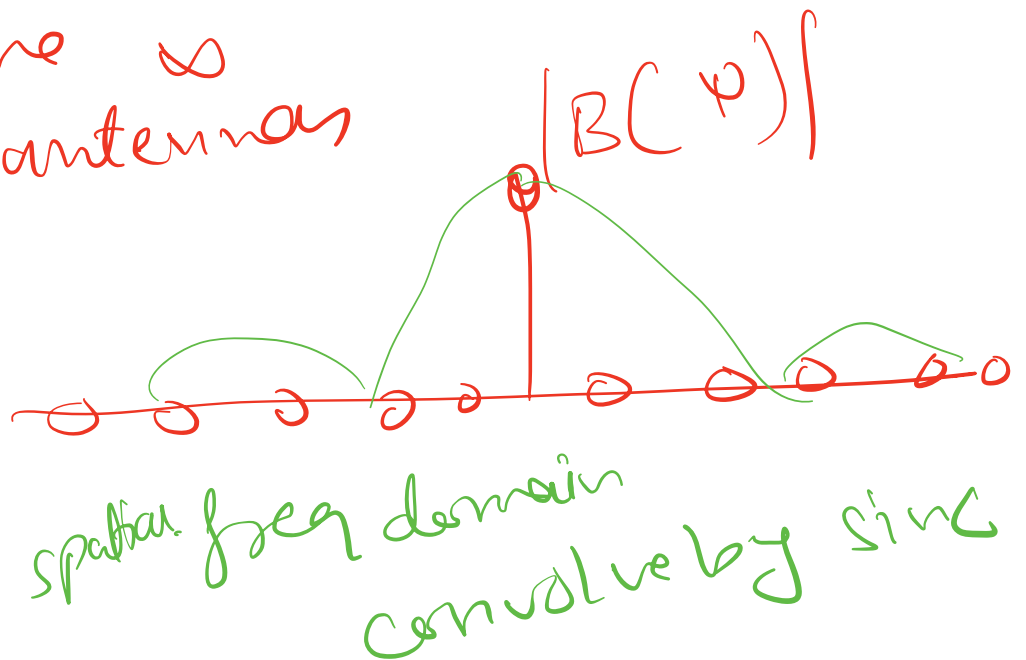
$$\underline{B(\psi)} = \sum_K h_K e^{-j 2\pi K \underbrace{\cos \psi}_{\text{freq}}}$$

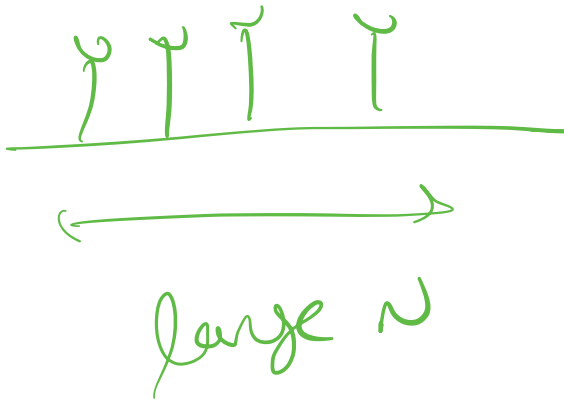
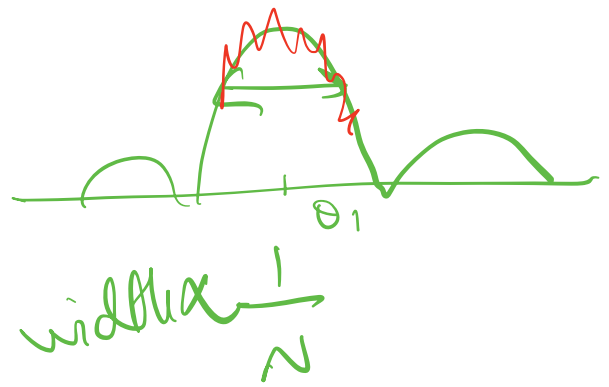
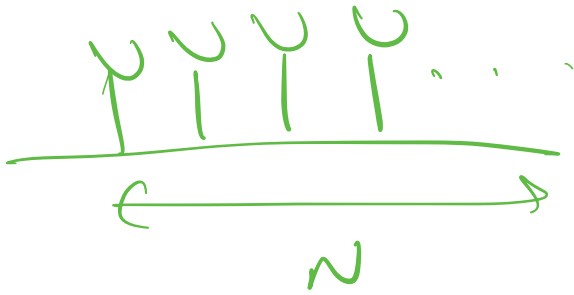
ψ is a variable
 θ_1 is a constant



if you have ∞
 antennas

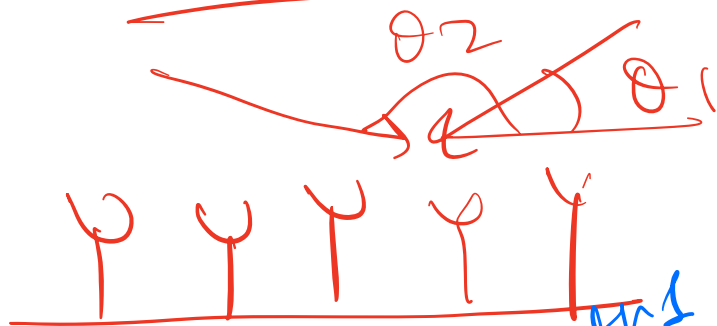
time
 domain
 multiply by
 a rect
 window





$\Rightarrow$ more accurate estimate of $\hat{\theta}_1$

with multipath



angle of arrival of path 1

$$K \pi \cos \theta_1$$

$$h_k \propto h_1 e$$

$$K \pi \cos \theta_2$$

K th antenna

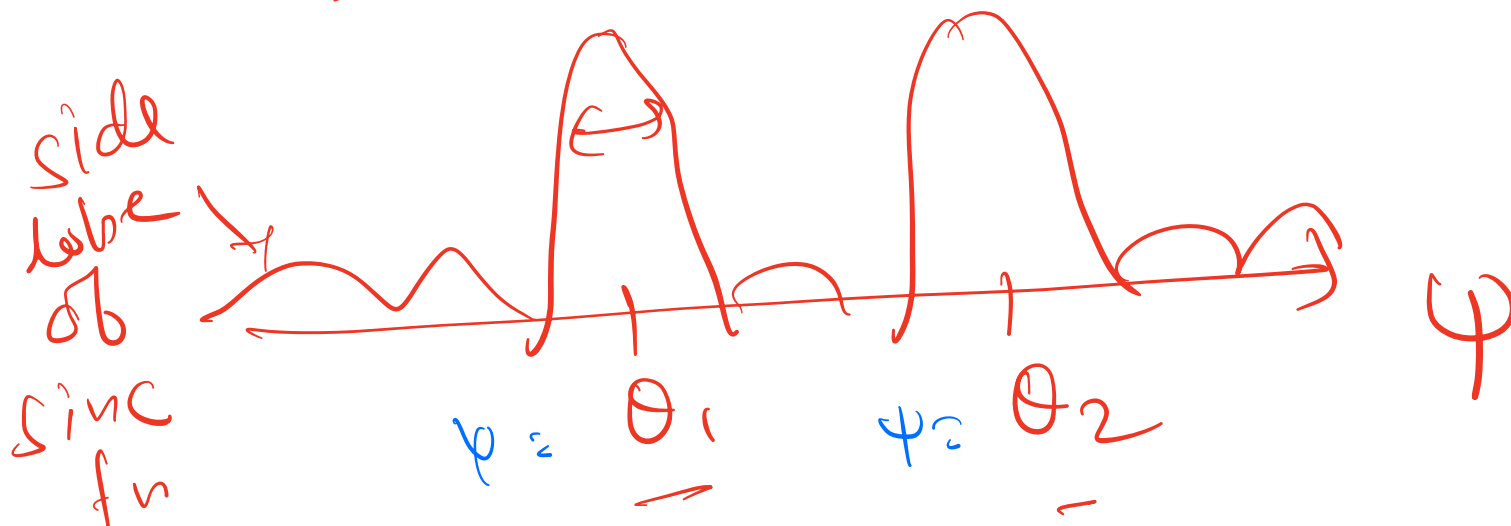
antenna 1

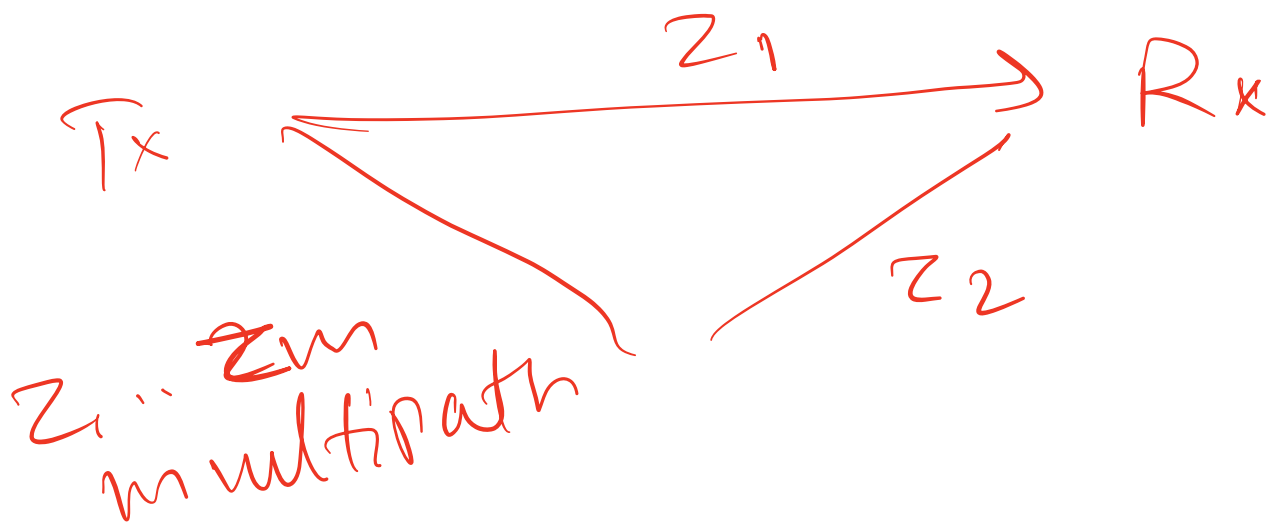
$$+ h_2 e$$

channel at antenna 1 path 2

angle of path 2

$$B(\psi) = \text{fft}(h_k)$$





$$h = \sum \delta(t - z_m)$$

baseband

$$y(f) \xleftarrow{\text{filter}} h(f) x(f) + n$$

$$y(t) = h(t) * h(t)$$

$$h(f) \propto \sum_m e^{j 2\pi f_c z_m}$$

$$\lambda = c/f$$

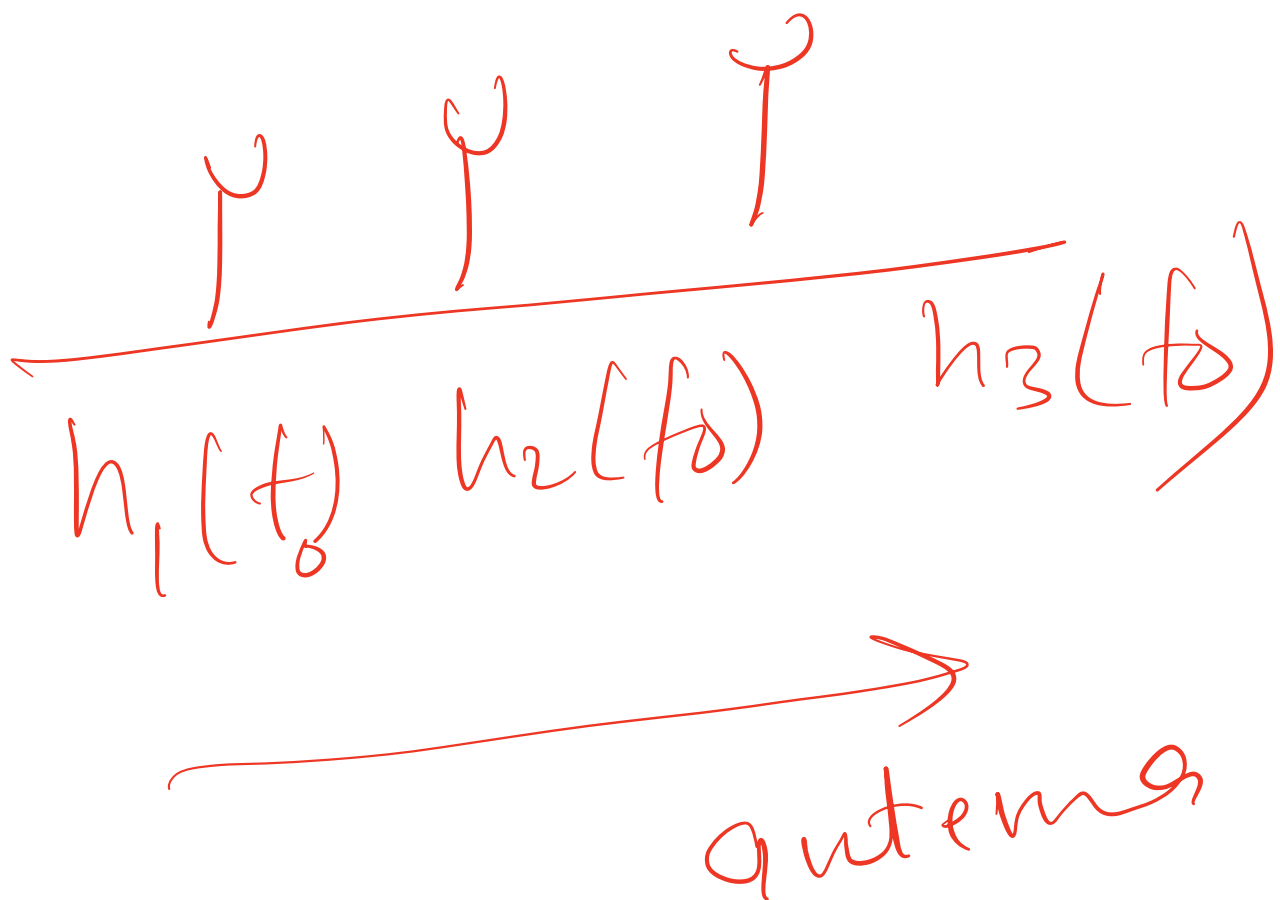
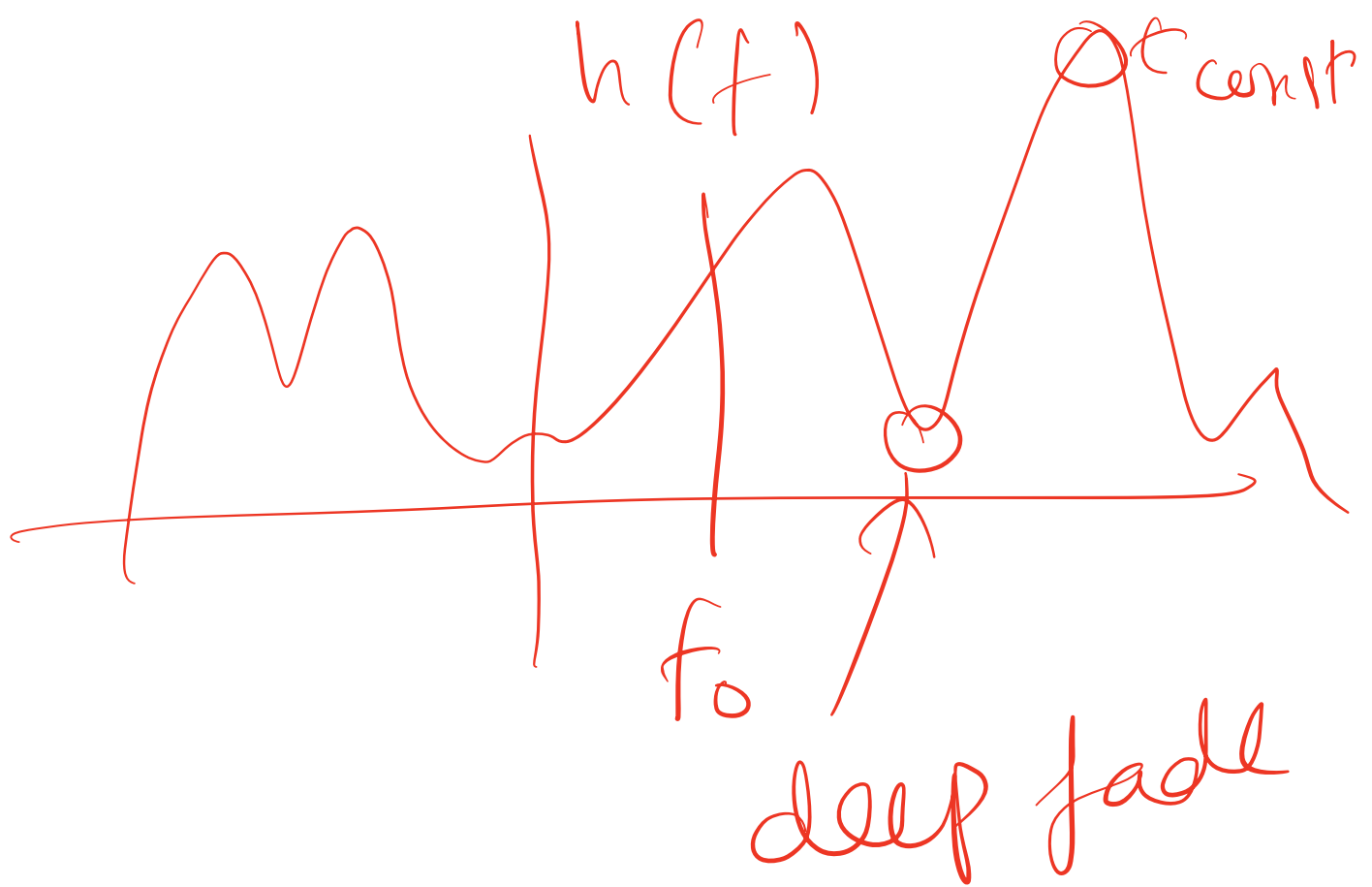
$$h(f) = \sum_m \frac{A}{d_m} e^{j2\pi f z_m}$$

$$f = \frac{c}{\lambda}$$

$$z_m = \frac{d_m}{c}$$

$$h(f) = \sum_m \frac{A}{d_m} e^{j2\pi f \frac{d_m}{c}}$$

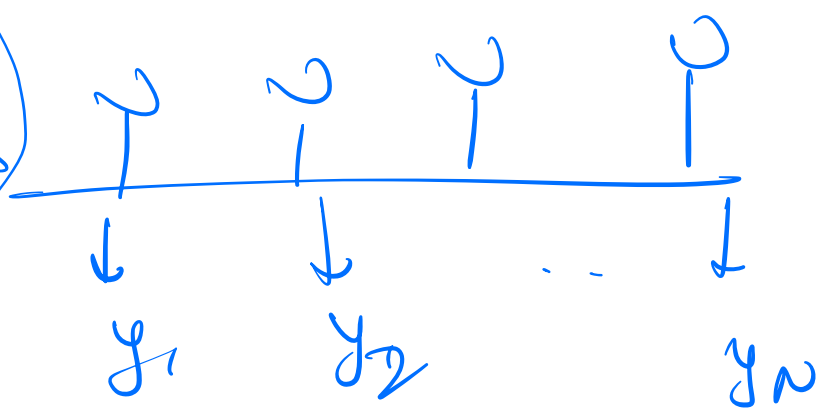
$$= \sum_m \frac{A}{d_m} e^{j2\pi \frac{d_m}{\lambda}}$$



transmit pilots

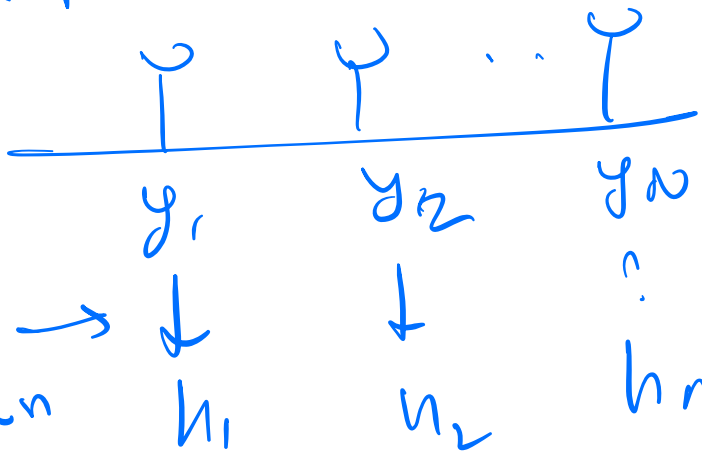
Step 1

mimo
Receive
diversity



receive
signal
at N
antennas

Step 2



Pilots
x is
known
to
the
RX

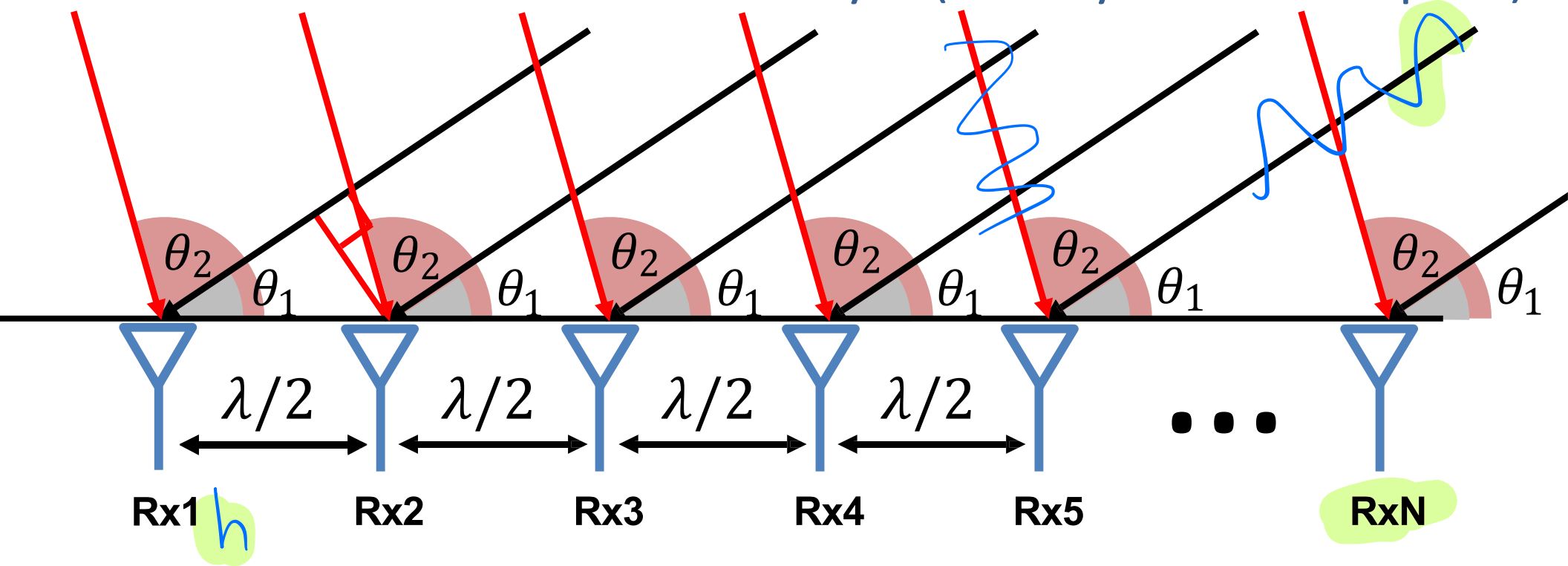
estimate
channel
noise
each
antenna

Step 3

Take FFT across $h_1, \dots, h_N$
give you $B(f)$

Step 4 : find peaks in $B(f)$
 $\Rightarrow$ estimated
AOA θ_1, θ_2

Method 3: Antenna Arrays (ArrayTrack Paper)

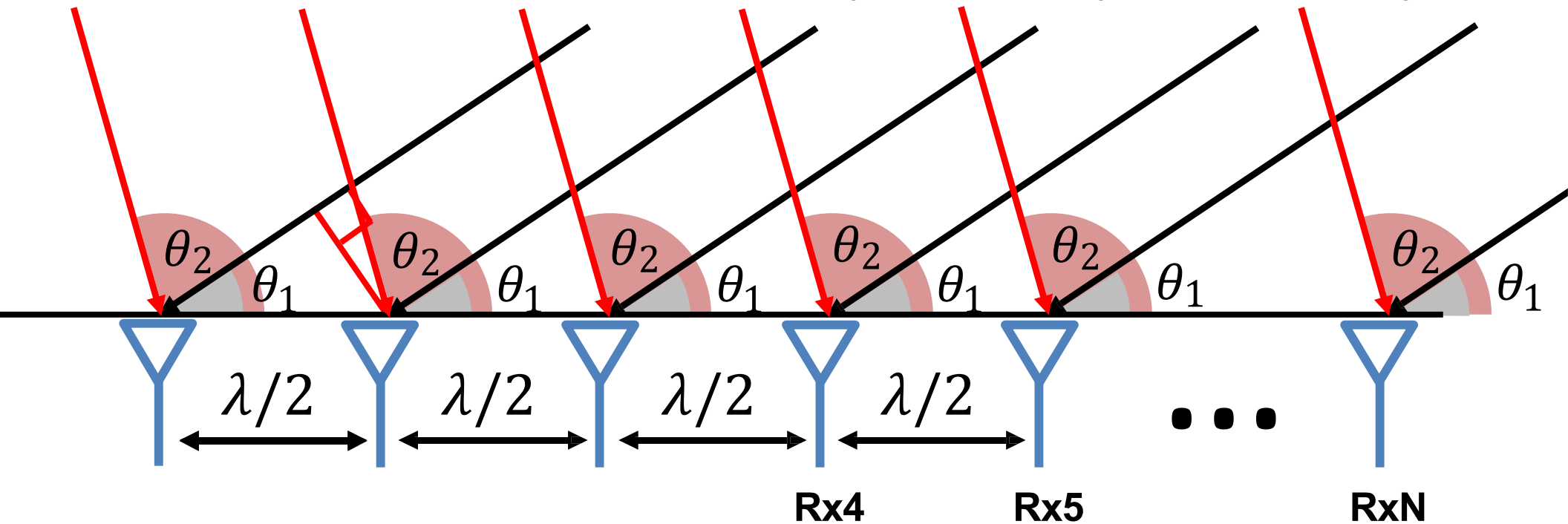


$$h_k = \alpha_1 e^{-j2\pi \frac{d_1 - k s \cos \theta_1}{\lambda}} = \alpha_1 e^{-j\phi_1 + k\pi \cos \theta_1} \\ + \alpha_2 e^{-j2\pi \frac{d_2 - k s \cos \theta_2}{\lambda}} + \alpha_2 e^{-j\phi_2 + k\pi \cos \theta_2}$$

For L paths $\rightarrow$

$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

Method 3: Antenna Arrays (ArrayTrack Paper)

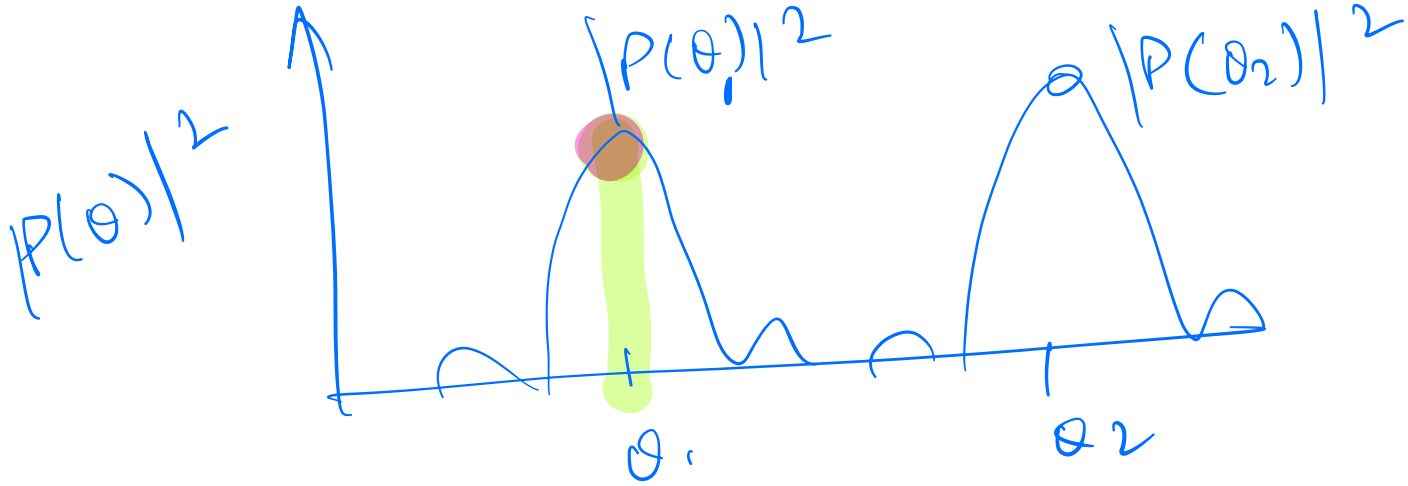


$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

$\beta(\psi)$ in the notes
or ψ

Multipath Profile: $P(\theta) = \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2$

Handwritten notes: $\beta(\psi)$ in the notes or ψ . A blue arrow points from the $\beta(\psi)$ note to the h_k term in the equation. A blue bracket under the summation is labeled $1-\tau$.



Method 3: Antenna Arrays (ArrayTrack Paper)

$$e^{-j2\pi f t}$$

$$P(\theta) = \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2$$

$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

$$P(\theta_1) = \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta_1} \right|^2$$

$$P(\theta_2) \approx N^2 \alpha_1^2$$

$$= \left| \sum_{k=1}^N \left(\sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l} \right) e^{-jk\pi \cos \theta_1} \right|^2$$

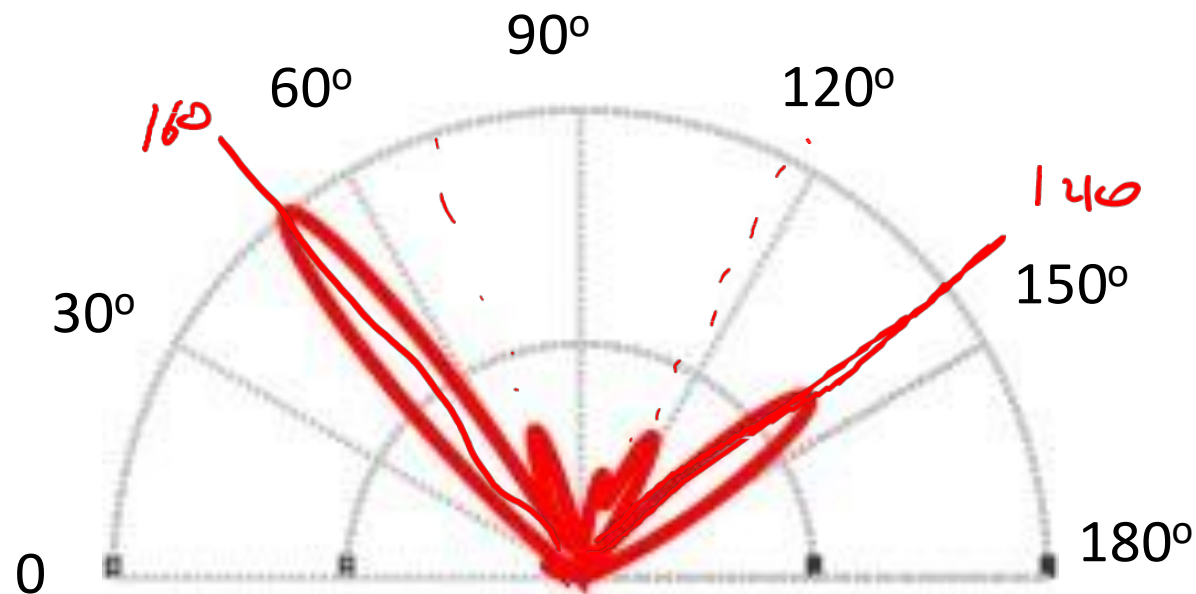
$$= \left| \sum_{k=1}^N \alpha_1 e^{-j\phi_1} + \sum_{k=1}^N \left(\sum_{l=2}^L \alpha_l e^{jk\pi (\cos \theta_l - \cos \theta_1) - j\phi_l} \right) \right|^2$$

$$= \left| N\alpha_1 e^{-j\phi_1} + \sum_{l=2}^L \alpha_l e^{-j\phi_l} \left(\sum_{k=1}^N e^{jk\pi (\cos \theta_l - \cos \theta_1)} \right) \right|^2 \approx N^2 \alpha_1^2$$

Method 3: Antenna Arrays (ArrayTrack Paper)

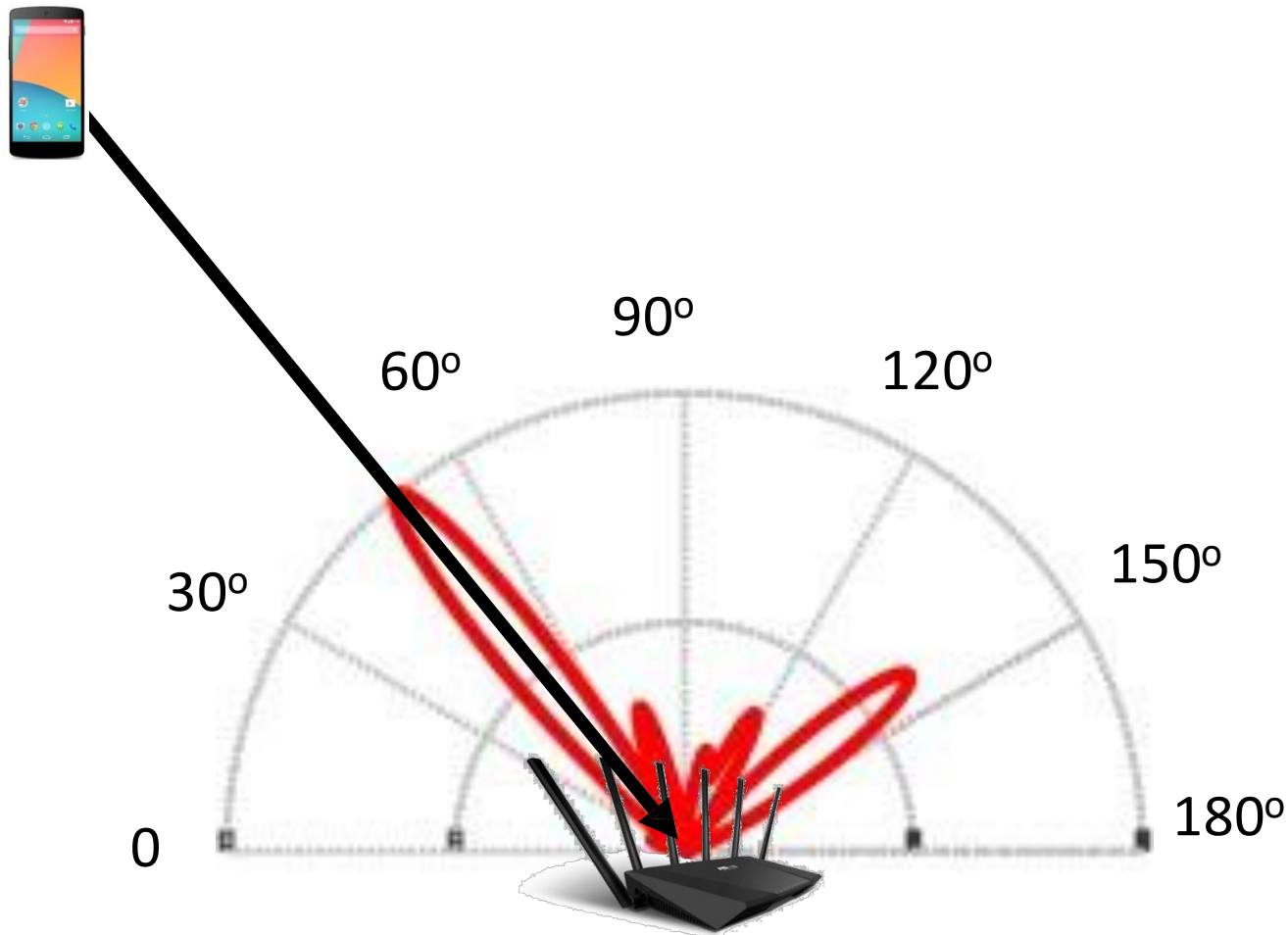
$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

$$\text{Multipath Profile: } P(\theta) = \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2$$



Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?
Strongest Path!

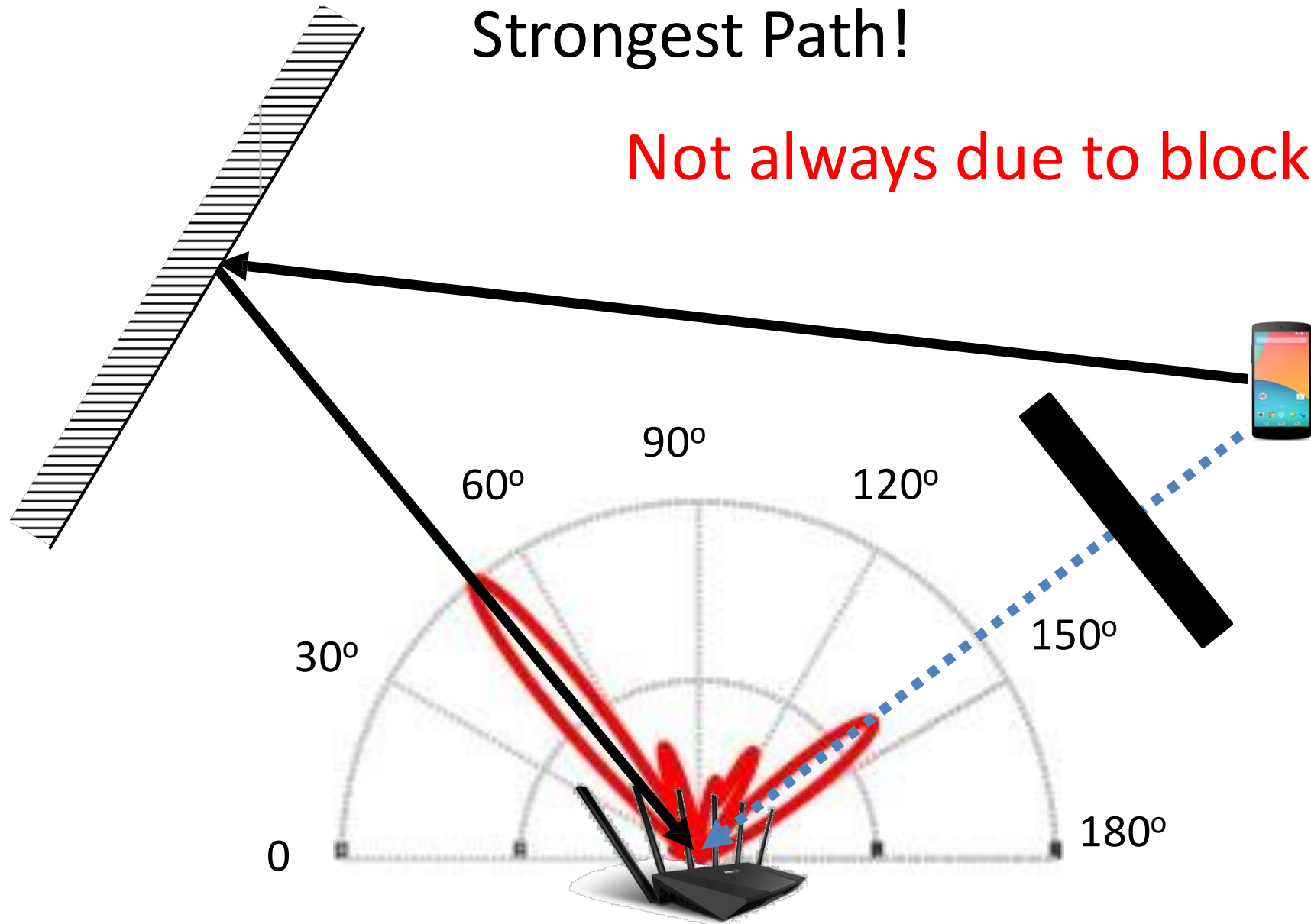


Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?

Strongest Path!

Not always due to blockage!

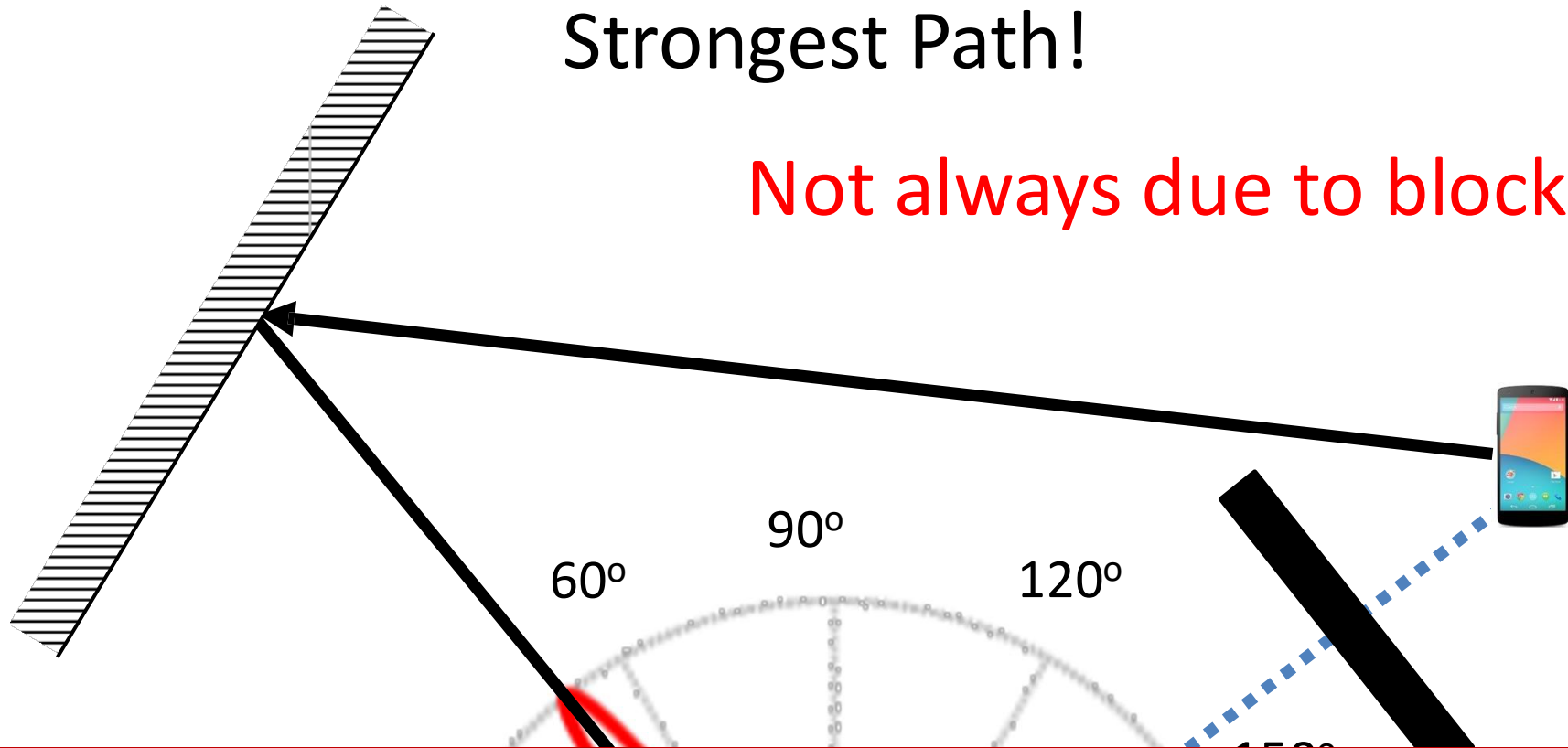


Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?

Strongest Path!

Not always due to blockage!



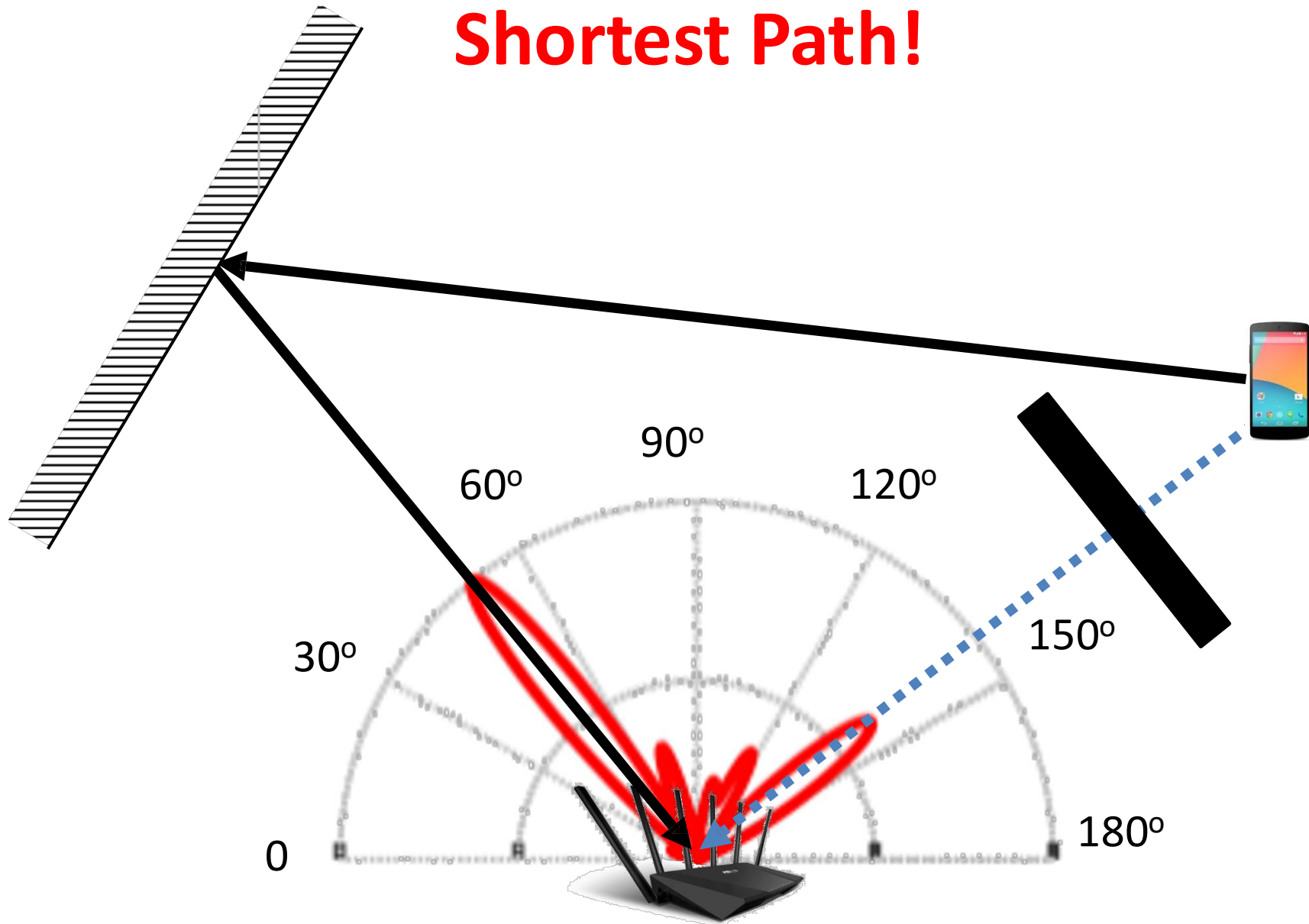
ArrayTrack: Leverage Mobility

- Line of sight path relatively stable with mobility
- Multipath reflection changes faster with mobility

Method 3: Antenna Arrays

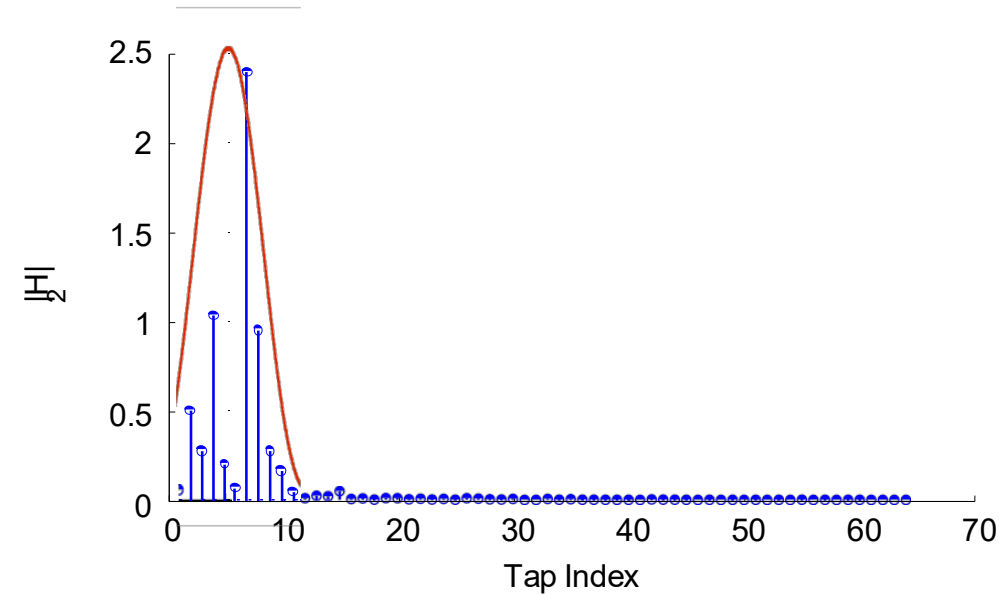
Which is the Line-of-Sight Path (Direct Path)?

Shortest Path!



Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



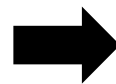
Multipath Profile vs Time

$$\Delta r = \frac{\Delta d}{c}$$

$$\Delta d = 1m$$

$$c = 3 \times 10^8 m/s$$

$$\Delta r = 3.3ns$$



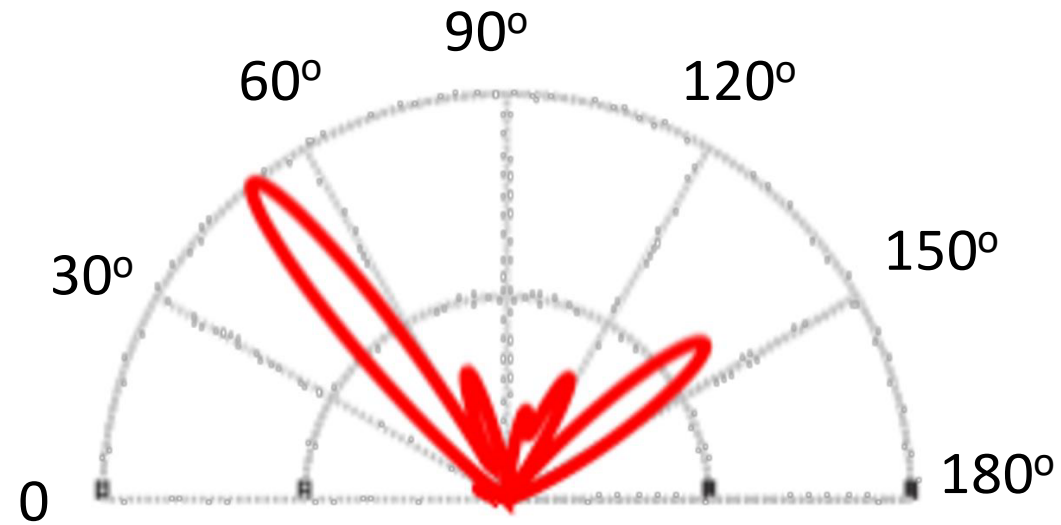
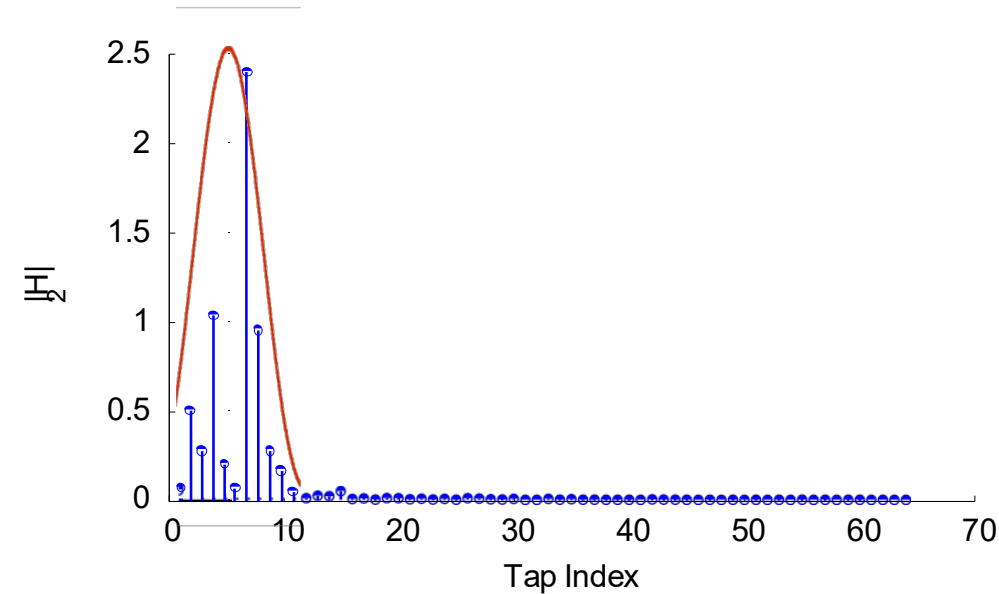
Requires a sampling rate

$$1/\Delta r = 300MHz$$

802.11n bandwidth = 40MHz

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

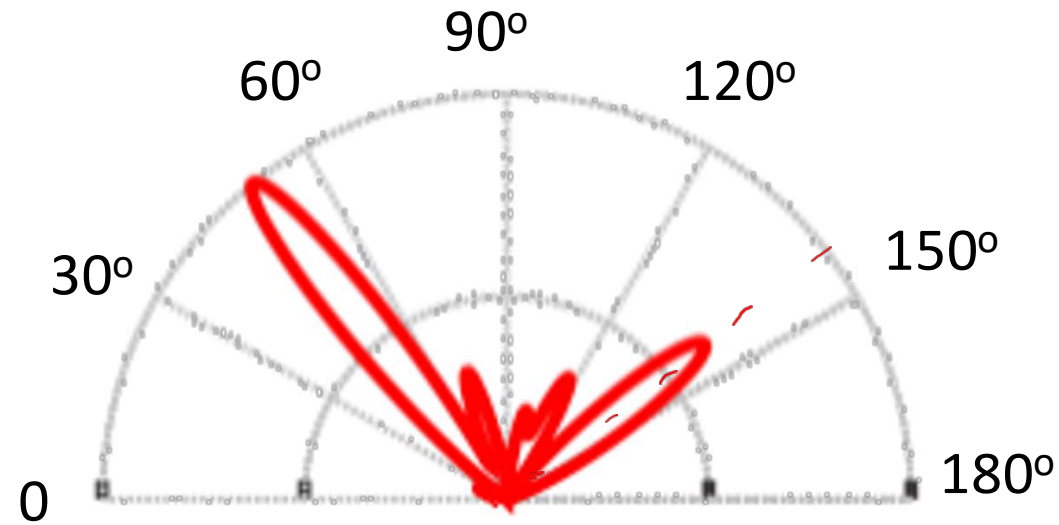
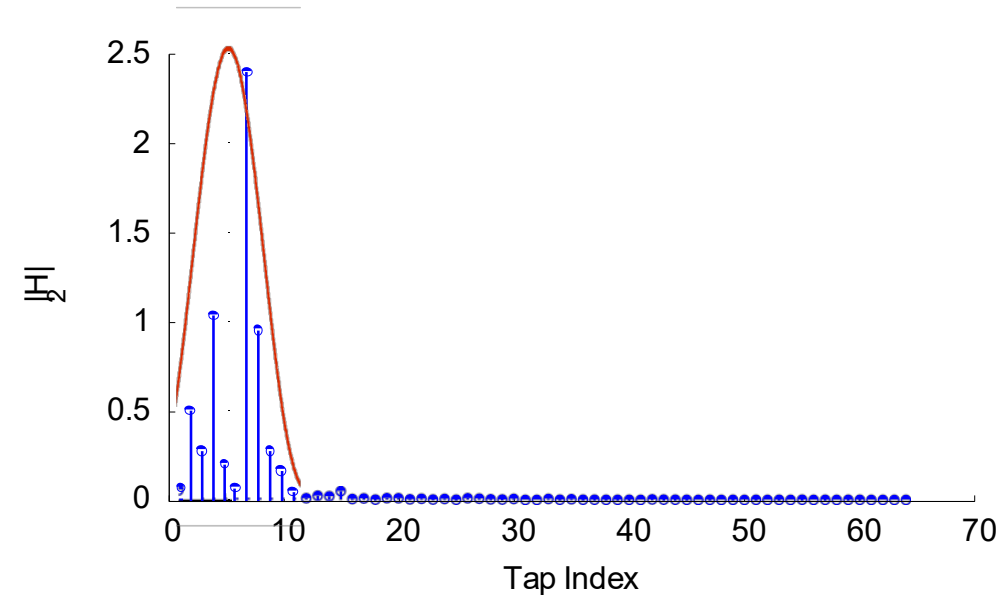
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path
3. Find the shortest path

But How?

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

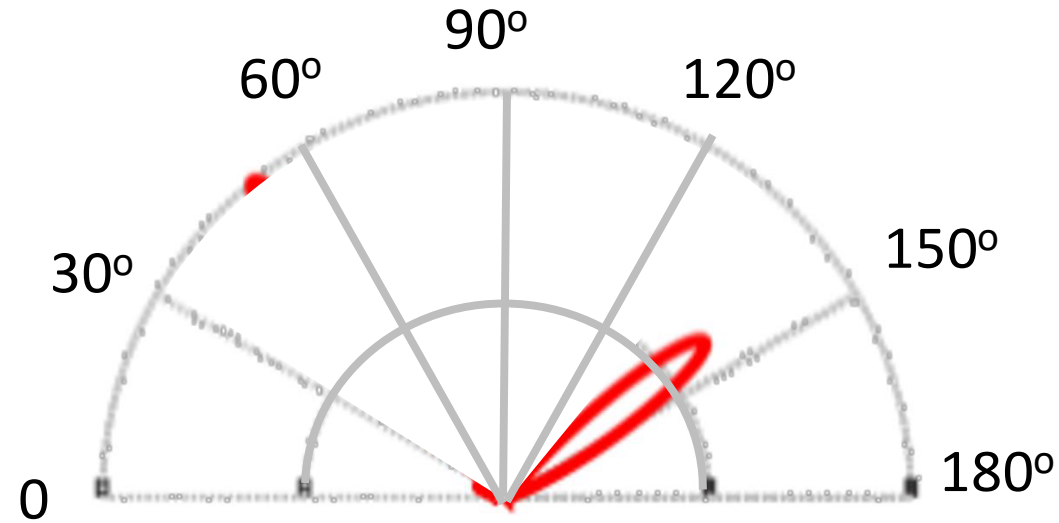
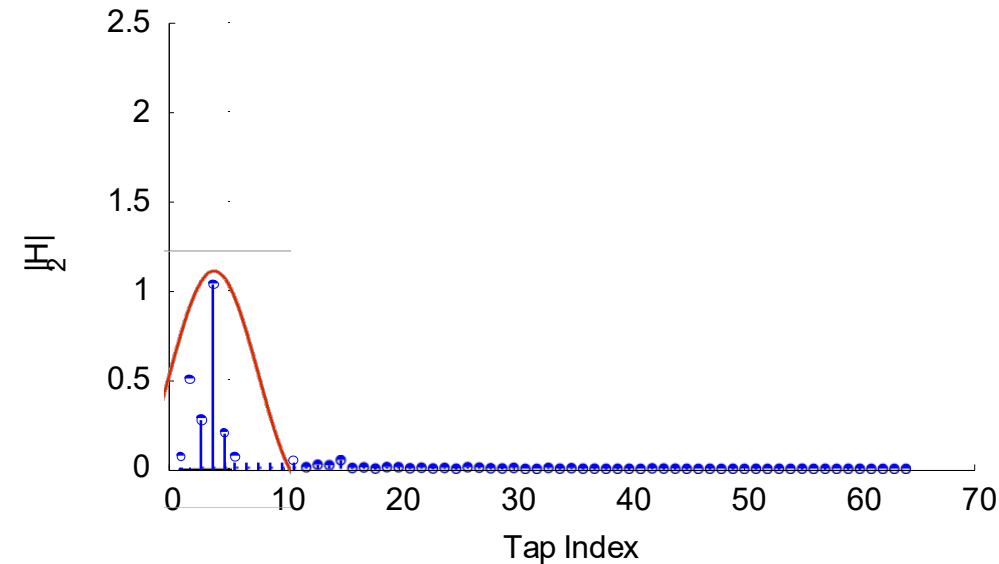
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths

$$y_{\theta_1}(t) = \sum_{k=1}^N y(t) e^{-jk\pi \cos \theta_1}$$

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

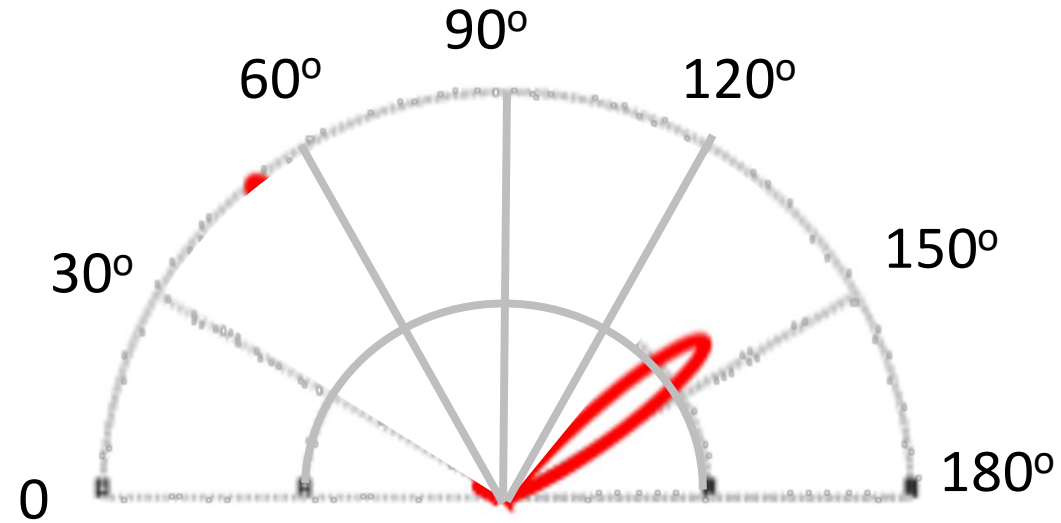
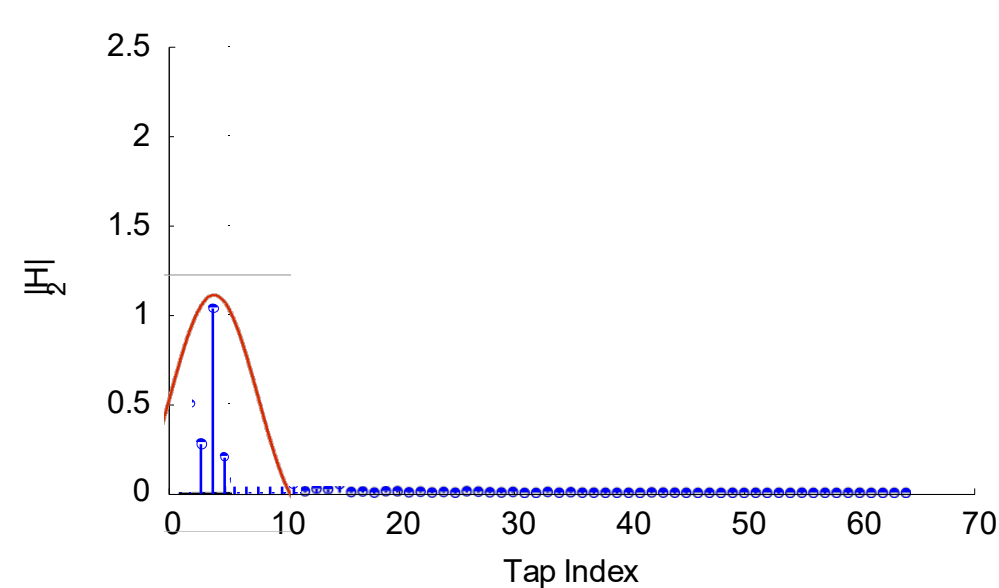
Multipath Profile vs AoA

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Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

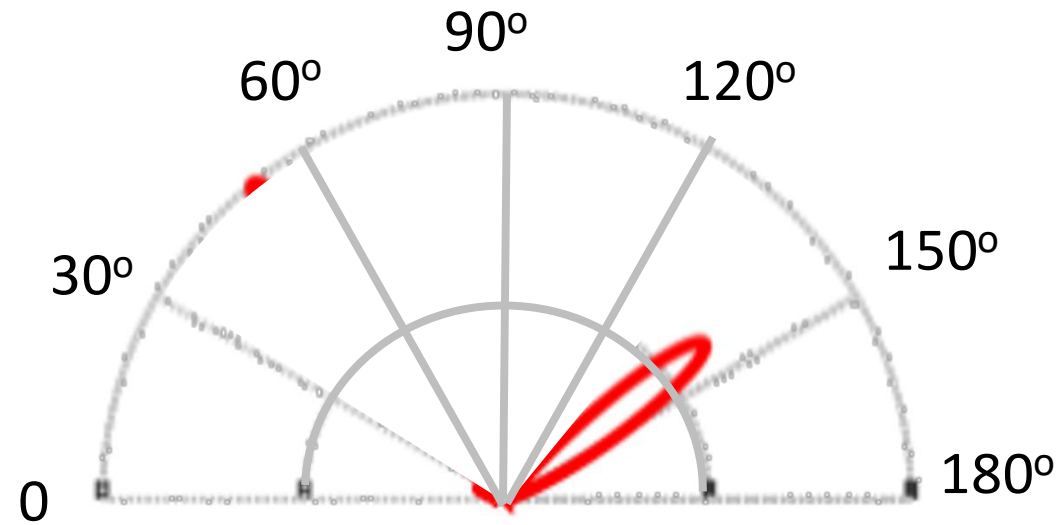
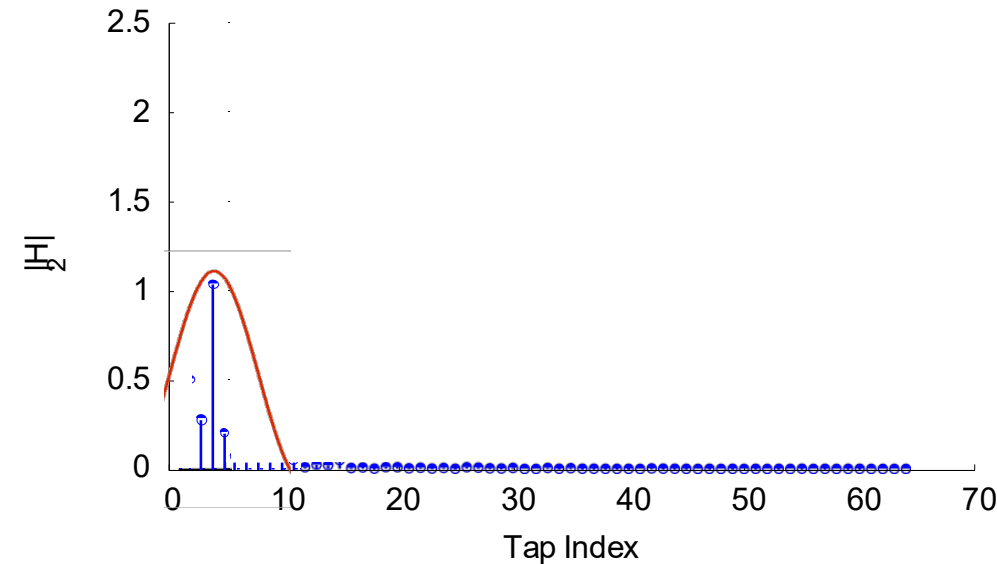
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path

Time Resolution still not enough ➡ Use OFDM

Method 3: Antenna Arrays

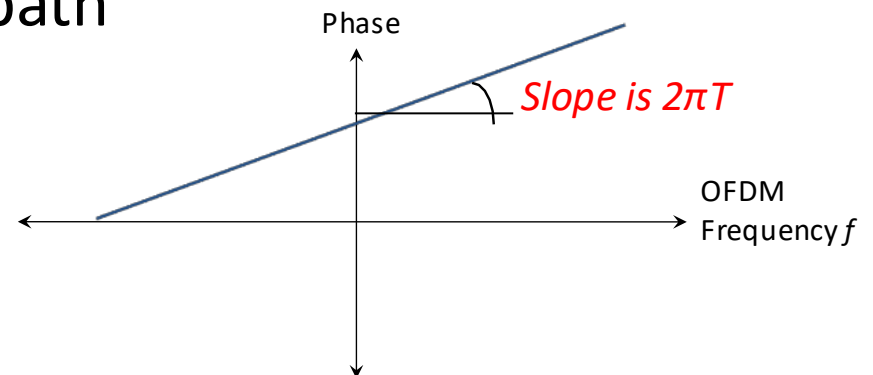
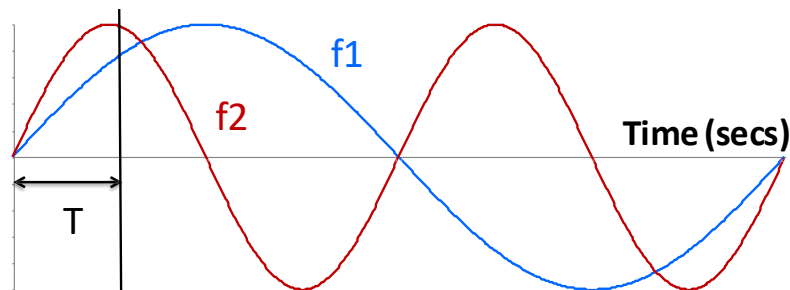
Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

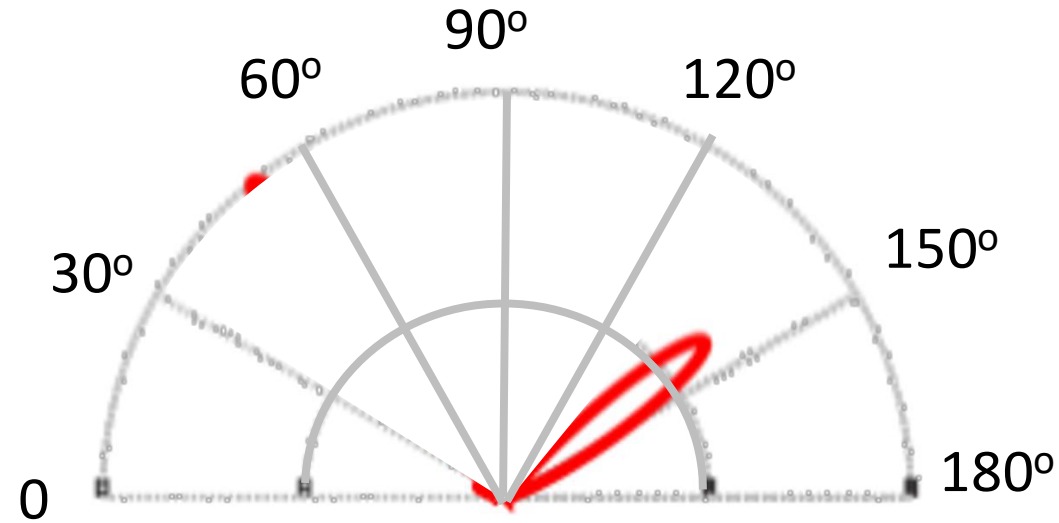
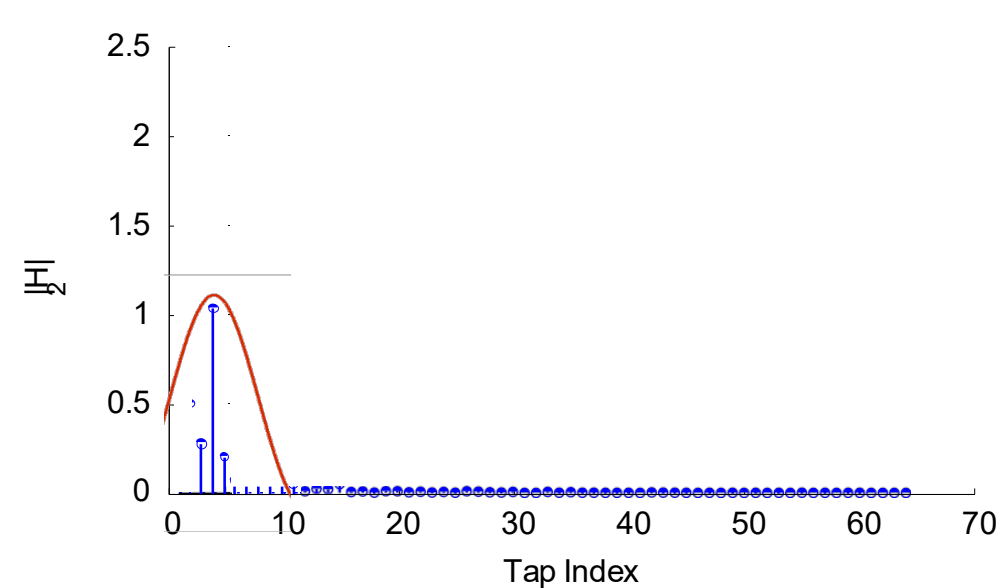
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path



Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

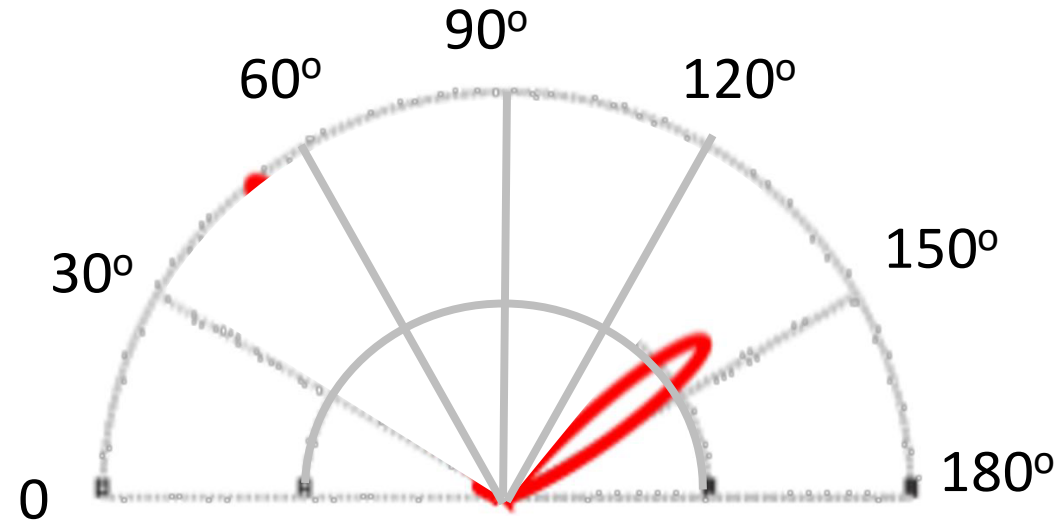
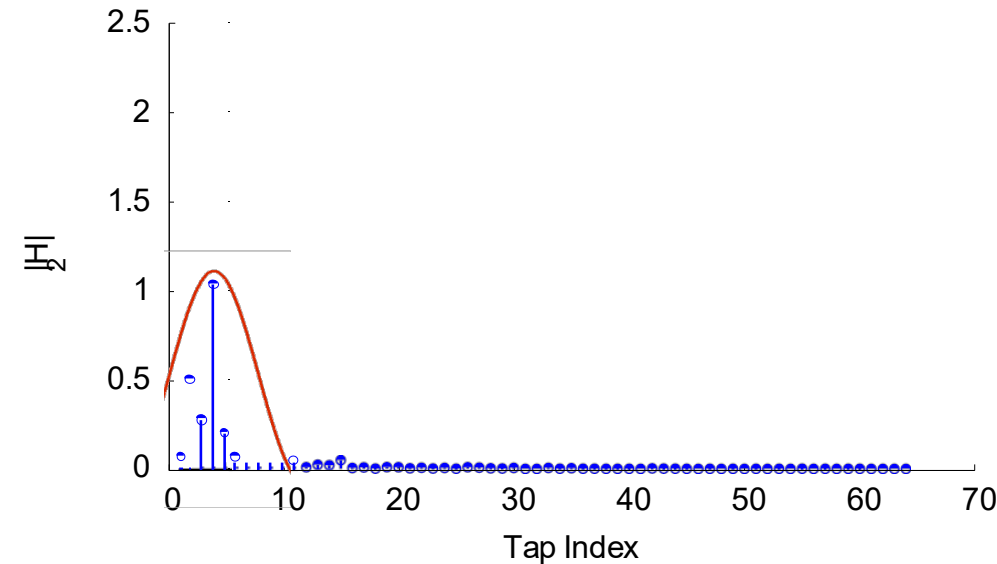
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path
 - Use OFDM to estimate delay from slope of phase vs freq.

But this delay includes packet detection delay & processing delay, not just propagation delay!

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?

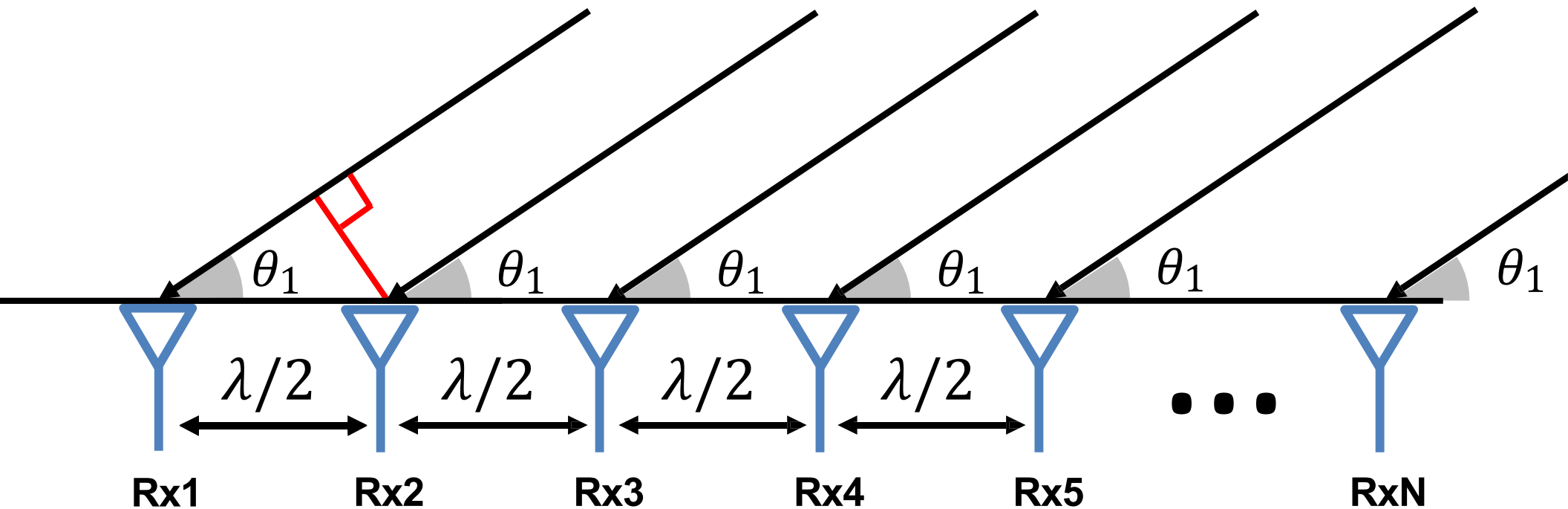


Multipath Profile vs Time

Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path
 - Use OFDM to estimate delay from slope of phase vs freq.
 - Compute relative delay for different paths!
3. Compare relative delays to find the shortest path

Method 3: Antenna Arrays



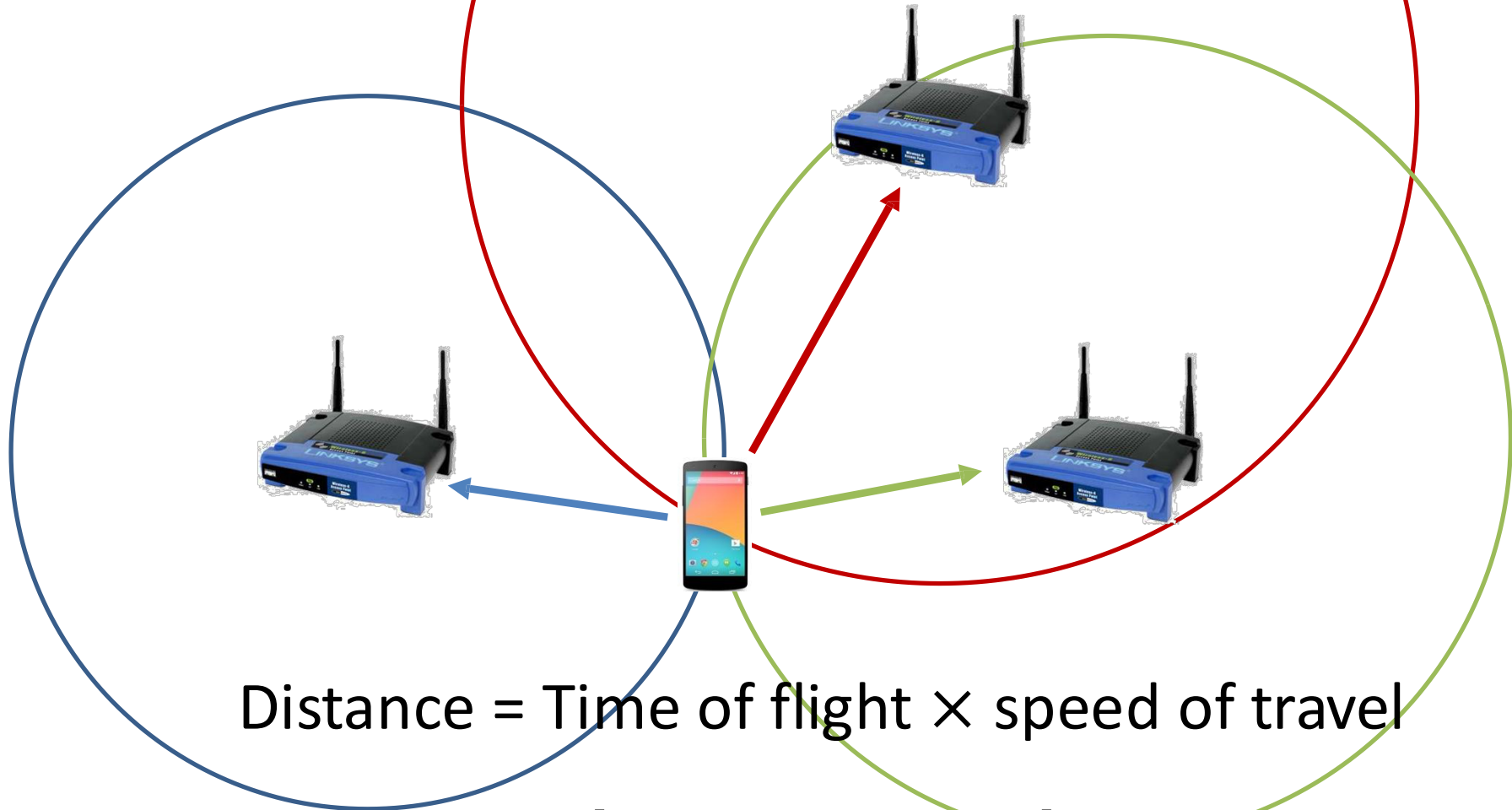
Pros: Works with multipath, No need for fingerprinting

Cons: Requires more hardware!

Assumes device is sufficiently far such that
wavefront is parallel

Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP



Distance = Time of flight $\times$ speed of travel

Measure ToF $\rightarrow$ Get distance $\rightarrow$ Trilateration

Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

Challenges:

- How do you know when signal was transmitted?



Measure Round Trip

- How about packet detection delay & processing delay?
 - Use OFDM to correct for packet detection delay
 - Estimate and calibrate for processing delay

Not Practical!

Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

Challenges:

- How do you know when signal was transmitted?



Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

Challenges:

- How do you know when signal was transmitted?



Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

Challenges:

- Accuracy limited by sampling rate (bandwidth)!

$$\Delta d = \Delta \tau \times c$$

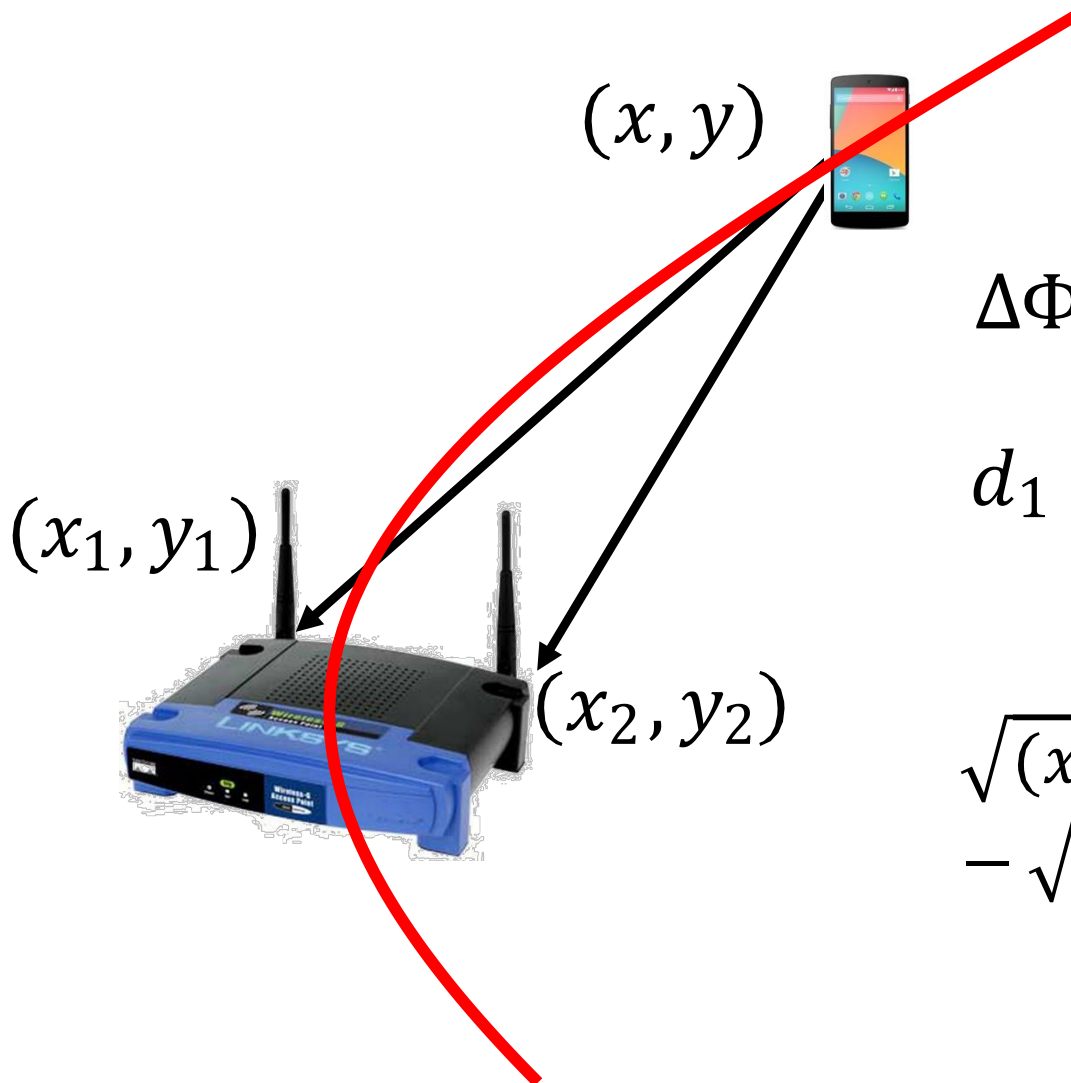
802.11n bandwidth = $40MHz$ $\Rightarrow \Delta r = 25ns$ $\Rightarrow \Delta d = 12.5\ m$

- Other systems than WiFi can get accurate ToF:
 - UWB: Ultra-Wide Band
 - FMCW: Frequency Modulated Carrier Wave

Not Supported in WiFi
(Will discuss in future lectures)

Method 5: TDoA Based Localization

Measure Time Difference of Arrival (TDoA) from device to AP's antennas



$$h_1 \propto e^{-j2\pi\frac{d_1}{\lambda}} \quad h_2 \propto e^{-j2\pi\frac{d_2}{\lambda}}$$

$$\Delta\Phi = \angle h_2 - \angle h_1$$

$$d_1 - d_2 = \lambda\Delta\Phi/2\pi = d'$$

Equation of Hyperbola

$$\sqrt{(x - x_1)^2 + (y - y_1)^2} - \sqrt{(x - x_2)^2 + (y - y_2)^2} = d'$$

Method 5: TDoA Based Localization

Measure Time Difference of Arrival (TDoA) from device to AP's antennas

